THE MAXIMUM OF FRACTIONAL BROWNIAN MOTION AS $H \to 0$: SHARP ASYMPTOTICS, FREE ENERGY, AND A CONTINUOUS RANDOM ENERGY MODEL

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ABSTRACT. Let M_H be the expected supremum on $[0,1]$ of a fractional Brownian motion with Hurst index H. We prove that $\sqrt{H}\, M_H \to \sqrt{2}$ as $H \downarrow 0$, which answers a question of Bisewski; it was previously known only that the limit points lie in $[1.128, 1.695]$. The upper bound $M_H \le \sqrt{2/H} + \sqrt{H}(2 + \log(1/H))$ holds for every $H \le \frac{1}{2}$. The mechanism is that, as $H \downarrow 0$, fractional Brownian motion behaves like a continuous random energy model with about $1/H$ effective scales, each carrying variance of order H. We make this precise by two-sided Gaussian comparisons with b-adic tree majorants and Cantor-type tree minorants. The method applies to any family of Gaussian processes on $[0,1]^d$ whose variogram at small scales is close to $\Phi(\varepsilon \log(1/|t-s|))$ for a convex profile Φ. For such families the expected supremum and the free energy are asymptotically given by the Bovier–Kurkova formulas for the continuous random energy model. Examples include Lévy fractional Brownian fields, Riemann–Liouville processes and stationary processes with powered-exponential covariance. The limiting free energy has a third-order phase transition when $\Phi''(0) > 0$, at $\beta_c = \sqrt{2}$ for fractional Brownian motion. We also determine the limit of the maximum over the uniform n-point grid when $H \log n \to \kappa$; in particular, uniform grids with $\log n = O(1/H)$ points capture only a fraction of M_H that stays bounded away from one.

CONTENTS

1. Introduction 2
2. Statement of results 5
3. Preliminaries 11
4. Upper bounds: tree majorants and a barrier lemma 14
5. Lower bounds: Cantor minorants 19
6. Proofs of the main theorems 23
7. Maxima over grids 25
8. Numerical illustration 29
9. Discussion and open problems 32
References 34

Date: October 2026.
2020 *Mathematics Subject Classification.* Primary 60G22, 60G15, 60G70; Secondary 60G60, 82B44, 60J80.
Key words and phrases. fractional Brownian motion, small Hurst index, expected supremum, Gaussian comparison, Sudakov–Fernique inequality, generalised random energy model, branching random walk, free energy, phase transition, Pickands' constant, log-correlated fields.

[†]ETH Zürich has not endorsed this work, and the work does not represent ETH Zürich. The affiliation is stated for identification purposes only. All opinions expressed are the author's own, and the author holds full responsibility for the content of this paper.

1. Introduction

1.1. **The problem.** A fractional Brownian motion (fBm) with Hurst index $H \in (0,1)$ is a centred Gaussian process $B^H = (B_t^H)_{t\geq 0}$ with $B_0^H = 0$ and

$$\mathbb{E}\big(B_t^H - B_s^H\big)^2 = |t-s|^{2H}, \qquad s,t \geq 0. \tag{1.1}$$

Up to a multiplicative constant, which (1.1) fixes, it is the unique centred Gaussian process that is H-self-similar with stationary increments [55, 62]. Since $\mathbb{E}(B_t^H)^2 = t^{2H}$ and $\mathbb{E}(B_t^H - B_s^H)^2 \leq 1$ on $[0,1]$, the process becomes increasingly irregular as $H \downarrow 0$ while its marginal variances stay bounded. Its supremum nevertheless diverges. The purpose of this paper is to determine the precise rate at which the expected supremum

$$M_H := \mathbb{E} \sup_{t\in[0,1]} B_t^H$$

diverges, and to explain the mechanism behind it.

For Brownian motion ($H = \frac{1}{2}$) the reflection principle gives $M_{1/2} = \mathbb{E}|B_1^{1/2}| = \sqrt{2/\pi}$. For no other value of $H \in (0,1)$ is a closed form known, and a sizeable literature is devoted to bounds, expansions and numerical schemes for M_H: see Delorme and Wiese [31] for a perturbative expansion around $H = \frac{1}{2}$, Bisewski, Dębicki and Rolski [9] for derivatives in H, and [13, 14, 8, 7, 59, 15] for bounds and simulation. In the regime $H \downarrow 0$ the order of magnitude $H^{-1/2}$ was identified by Borovkov, Mishura, Novikov and Zhitlukhin [13], who proved $(4\pi e H \log 2)^{-1/2} \leq M_H \leq 16.3\, H^{-1/2}$. The upper constant was later improved to 1.695 [14, Cor. 2 and Rem. 5], and the lower one, via an explicit coupling, to $2/\sqrt{\pi} \approx 1.128$ by Bisewski [7, Cor. 3]:

$$1.128 \;\leq\; \liminf_{H\downarrow 0} \sqrt{H}\, M_H \;\leq\; \limsup_{H\downarrow 0} \sqrt{H}\, M_H \;\leq\; 1.695. \tag{1.2}$$

In [7, Section 3.1] it is stated that determining whether $\lim_{H\downarrow 0} \sqrt{H} M_H$ exists, and finding its value, "remains an interesting open question". Our first result answers this question.

Theorem 1.1. *As $H \downarrow 0$,*

$$\sqrt{H}\, M_H \longrightarrow \sqrt{2}.$$

More precisely, for every $H \in (0, \frac{1}{2}]$,

$$M_H \leq \sqrt{2/H} + \sqrt{H}\,\big(2 + \log(1/H)\big), \tag{1.3}$$

and for every $H \in (0, 3\cdot 10^{-6}]$,

$$M_H \geq \sqrt{2/H} - 2\sqrt{2}\,\sqrt{\log(1/H)}. \tag{1.4}$$

Moreover $\mathbb{P}(|\sup_{[0,1]} B^H - M_H| > u) \leq 2e^{-u^2/2}$ for all $u > 0$, so that $\sqrt{H}\sup_{[0,1]} B^H \to \sqrt{2}$ in L^p for every $p < \infty$.

The constant $\sqrt{2} \approx 1.414$ lies strictly inside the window (1.2), and the error in the upper bound (1.3) *vanishes* as $H \downarrow 0$. The lower bound (1.4) is the closed form of a bound that holds for every $H \in (0,1)$, namely Theorem 2.1(ii) below. Evaluated numerically (Table 1), our upper bound improves on that of [14] for all $H \leq 0.24$, and our lower bound improves on that of [7] for $H \leq 0.01$. Theorem 1.1 is a special case of Theorem 2.1.

A recent conjecture of Mukeru and Mulaudzi [64], $M_H = 1/(H\sqrt{2\pi})$ for $H \in (0,1]$, is exact for $H = \frac{1}{2}$ and $H = 1$. It grows like H^{-1} as $H \downarrow 0$, and it violates (1.3) for every $H \leq 0.055$.

1.2. **Where $\sqrt{2}$ comes from.** Classical tools for suprema of Gaussian processes cannot detect the constant in Theorem 1.1. Dudley's entropy integral, Sudakov's minoration and Talagrand's generic chaining [41, 73, 75] determine $\mathbb{E}\sup$ only up to universal multiplicative constants. A union bound over a fine grid, on the other hand, ignores the correlation structure and produces the wrong rate. The constant is instead governed by a hierarchical structure that becomes exact as $H \downarrow 0$. To see it, write distances on a logarithmic scale: if $|t-s| = e^{-x}$ then

$$\mathbb{E}\big(B_t^H - B_s^H\big)^2 = e^{-2Hx}.$$

Three features of this formula drive everything that follows.

(a) *A common mode carries half the variance.* For fixed $s \neq t$ in $(0,1]$, $\mathrm{Cov}(B_s^H, B_t^H) \to \frac{1}{2}$ while $\mathrm{Var}(B_t^H) \to 1$. Half of the variance is shared by all points and has mean zero, so it cannot contribute to $\mathbb{E}\sup$.

(b) *The remaining variance is spread over about $1/H$ scales.* Between the log-scales x and $x+dx$, two points that "branch" acquire independent increments of variance $-\frac{1}{2}\frac{d}{dx}e^{-2Hx}\,dx = He^{-2Hx}\,dx$. The total is $\int_0^\infty He^{-2Hx}dx = \frac{1}{2}$, but this variance is spread over $\asymp 1/H$ units of log-scale, each carrying only $O(H)$.

(c) *Each unit of log-scale carries one unit of entropy.* There are e^x disjoint intervals of length e^{-x}.

In a branching random walk or a generalised random energy model (GREM) [33, 35, 20], a level with entropy E and variance a contributes $\sqrt{2\mathsf{a}\mathsf{E}}$ to the maximum, provided the variance per unit entropy is non-increasing along the hierarchy (a *concave* profile). Here the variance per unit entropy, He^{-2Hx}, is decreasing. Summing the contributions gives

$$M_H \approx \int_0^\infty \sqrt{2\cdot He^{-2Hx}}\,dx = \sqrt{2H}\cdot\frac{1}{H} = \sqrt{\frac{2}{H}}.$$

In the rescaled variable $y = Hx$ the fBm behaves, at the level of first-order limits, like a *continuous random energy model* (CREM) in the sense of Bovier and Kurkova [21], with variance profile $\Phi(y) = e^{-2y}$. The maximum of this CREM is $H^{-1/2}\int_0^\infty \sqrt{-\Phi'(y)}\,dy = \sqrt{2/H}$.

The heuristic also explains why simulation struggles. Observing B^H on n grid points resolves only the log-scales $x \le \log n$, so the grid maximum is close to $\sqrt{2/H}$ only if $H\log n \gg 1$, i.e. $n \ge e^{C/H}$. When $H\log n \to \kappa \le \frac{1}{2}$, the grid maximum is $\approx \sqrt{\log n}$, the value for n independent $N(0,\frac{1}{2})$ variables. Borovkov and Zhitlukhin [15] proved a normal approximation of this kind when $H(\log n)^2 \to 0$. Theorem 2.13 gives the full phase diagram in $\kappa \in [0,\infty]$.

1.3. **Strategy of the proofs.** For the upper bound we compare B^H with a GREM indexed by the b-adic intervals of $[0,1]$. The increments of the GREM dominate those of B^H at the corners of the intervals (Proposition 4.4), so the Sudakov–Fernique inequality bounds M_H by the expected maximum of the GREM. Its variance per unit entropy decreases along the tree, and an exponential-tilt barrier argument (Lemma 4.1) bounds its maximum non-asymptotically. The base b plays no role here to first order: for every b the b-adic GREM has first-order maximum $\sqrt{2/H}$ and the barrier bound is sharp for it, so $b = 2$ suffices.

The lower bound needs the comparison in the other direction. Points in different b-adic intervals can be arbitrarily close, so we pass to a Cantor-type subtree whose branches are separated by at least the size of their cells (Lemma 5.1). On this subtree the GREM increments are dominated by those of B^H. Because the profile is concave, the optimal path in the GREM takes an almost maximal increment at every level, so the greedy strategy, which picks at each level the child with the largest increment, is already sharp to first order (Proposition 5.3). The minorant does pay for its separated branches: the cell ratio $2b$ costs the entropy $\log 2$ at each

level, and the greedy bound replaces $\sqrt{2\log b}$ by μ_b. Both are relative losses of order at most $\log\log b/\log b$, so the lower bound needs $b \to \infty$. The common mode costs nothing in either direction, since both comparisons involve only increments.

For free energies the same two comparisons hold by the interpolation identity of Lemma 3.1. The upper bound uses the barrier on the levels where the GREM is frozen and a first moment on the others (Proposition 4.3), and the lower bound applies Jensen's inequality level by level (Proposition 5.4). Grids need one more ingredient. Below the grid spacing the effective profile lies below its chord, so the hierarchy is invisible there, and a truncated second moment argument shows that these scales behave like a single random energy model (Lemma 7.2).

1.4. **Main results.**

(i) Explicit two-sided non-asymptotic bounds for M_H, and the limit $\sqrt{H}M_H \to \sqrt{2}$ (Theorem 2.1).

(ii) The limiting free energy of $\int_0^1 \exp(\beta B_t^H/\sqrt{H})dt$, with a third-order phase transition at $\beta_c = \sqrt{2}$ (Theorem 2.2). In the parametrisation of Gaussian multiplicative chaos (Remark 2.3), β_c is the critical value $\sqrt{2}$ of one-dimensional chaos [52, 69]. Neuman and Rosenbaum showed that exponentials of a recentred version of $B^H/\sqrt{H}$ converge to this chaos in the subcritical range [65, Cor. 2.2].

(iii) A *universality principle* for families of Gaussian processes on $[0,1]^d$ whose variogram at small scales has the multiscale form $\Phi(\varepsilon\log(1/|t-s|))$ (Theorem 2.7 and Proposition 2.9). The limits are the Bovier–Kurkova CREM formulas (Remark 2.8). The principle covers Lévy fractional Brownian fields, Riemann–Liouville processes, mixtures of fBms and stationary fields with powered-exponential covariance (Corollary 2.11).

(iv) The phase diagram $\kappa \mapsto \Lambda(\kappa)$ of grid maxima, $\sqrt{H}\,\mathbb{E}\max_{i\le n} B^H_{i/n} \to \Lambda(\kappa)$ for $H\log n \to \kappa$ (Theorem 2.13), and the resulting first-order discretisation bias (Corollary 2.14).

(v) An improved asymptotic upper bound for Pickands' constants $\mathcal{H}_{2H}$ as $H \downarrow 0$ (Corollary 2.16).

In Section 9 we formulate conjectures on second-order corrections, on the fluctuations of the maximum and on overlaps. In particular we conjecture that $\sup B^H - M_H$ converges in law to $N(0,\frac{1}{2})$, the fluctuation being carried entirely by the common mode. The Monte Carlo data of Section 8 cannot test this conjecture: on feasible grids the centred grid maximum fluctuates like the maximum of independent variables (Section 8, item (d)).

1.5. **Related work and what is new.** The $H \downarrow 0$ limit of fBm has been studied from several angles. Fyodorov, Khoruzhenko and Simm [47] constructed a regularised "$H = 0$" fBm and related it to characteristic polynomials of GUE matrices. Neuman and Rosenbaum [65] showed that suitably renormalised fBm converges to a log-correlated Gaussian distribution as $H \downarrow 0$. Hager and Neuman [49], and Forde, Fukasawa, Gerhold and Smith [45], studied the associated multiplicative chaos. In [65, 49] the process is rescaled by a factor of order $H^{-1/2}$, as here, and recentred by a running mean, which removes the common mode of Section 1.2(a). These works concern *distributional* limits after renormalisation. Our results concern the sample-path extremes and the free energy of fBm itself, at the scale where its hierarchical structure is visible. Mönch [63] recently described the local structure of rough fBm at its maximiser for fixed H.

The mechanism of Section 1.2 is that of Derrida's GREM and its continuous version [32, 33, 35, 23, 20, 21]; see [16, 17] for expositions. Bovier and Kurkova [21] computed the maximum, the free energy and the overlap distribution of the CREM, and our limits are their formulas after a change of variables (Remark 2.8). Extracting such a hierarchy from a non-hierarchical field by Gaussian comparison is also an established approach. Comparison with tree-indexed

processes is a basic device for log-correlated fields [22, 40, 56, 39]; see [2, 10, 78] for expositions. Arguin and Zindy obtained the free energy and the overlap distribution of a log-correlated field on $[0,1]$ and of the two-dimensional discrete Gaussian free field by comparison with GREMs and perturbation of the variance across scales [5, 6]. For the Gaussian free field with scale-dependent variance, the maximum, the free energy and the overlaps were obtained in this way in [4, 66], and Arguin, Belius and Harper determined the maximum of a randomised model of the Riemann zeta function by extracting an approximate tree structure [3]. Time-inhomogeneous branching random walks and variable-speed branching Brownian motion are understood in great detail, including the case of decreasing variance [43, 42, 61, 58] and the two-speed and weak-correlation regimes [18, 19]. Kistler's notes [54] describe the underlying "multiscale refinement" of the second moment method.

The proof of Theorem 1.1 combines these standard tools once the comparison with the b-adic and Cantor trees is set up. Against this background, the new elements are the following.

(1) *The setting.* The processes are indexed by Euclidean cubes and are specified only through two-sided bounds on the variogram at small scales, with no hierarchical or lattice structure. The number of effective scales, of order $1/\varepsilon$, tends to infinity while each scale carries vanishing variance.
(2) *The Cantor minorant.* It produces a hierarchy with separated branches inside a Euclidean domain, and for convex profiles the greedy strategy on it is sharp to first order.
(3) *Explicit bounds.* The basic estimates (Lemmas 4.1 and 7.2 and Propositions 4.3, 5.3 and 5.4) are non-asymptotic, and for fBm they give explicit bounds that improve those of [14, 7] for small H.
(4) *Applications.* These are the answer to the question of [7], the free energy of fBm, and the grid phase diagram. The last of these requires a second moment estimate for sub-chord GREMs with a growing number of levels (Lemma 7.2).

Gaussian comparison is classical [73, 44, 53]; we use the interpolation approach of Chatterjee [25]. For Pickands' constants see [68, 72, 37, 50, 51].

1.6. **Organisation and notation.** Section 2 states the results. Section 3 collects comparison and concentration tools, bounds for Gaussian maxima and a one-level random energy estimate. Section 4 develops tree majorants and the barrier lemma, and Section 5 Cantor minorants. Section 6 proves Theorems 2.1, 2.2 and 2.7 and Proposition 2.9 and the corollaries, and Section 7 treats grids. Section 8 contains numerical illustrations, and Section 9 discusses open problems.

Throughout, $|\cdot|$ denotes the sup-norm on $\mathbb{R}^d$, $\mathsf{T} = [0,1]^d$, and for a centred Gaussian process X we write $\mathsf{d}_X(s,t) := (\mathbb{E}|X_t - X_s|^2)^{1/2}$. We use log for the natural logarithm, $\log^+ x = \max(\log x, 0)$, and $\mu_n := \mathbb{E}\max_{1\le i\le n}\xi_i$ for i.i.d. standard normal ξ_i (with $\mu_1 = 0$). All Gaussian processes are centred.

2. Statement of results

2.1. **Fractional Brownian motion.** Recall $M_H = \mathbb{E}\sup_{t\in[0,1]} B_t^H$ and $\mu_n = \mathbb{E}\max_{i\le n}\xi_i$.

Theorem 2.1 (Sharp asymptotics of M_H)**.**

(i) *For every $H \in (0,1)$, every integer $b \ge 2$ and every $\delta > 0$,*

$$M_H \le (1+\delta)\sqrt{\log b \,\coth\!\big(\tfrac{1}{2}H\log b\big)} + \frac{1+\log^+ P_{b,\delta}}{c_{b,\delta,H}}, \tag{2.1}$$

where $P_{b,\delta} := (b^{2\delta+\delta^2}-1)^{-1}$ and $c_{b,\delta,H} := (1+\delta)\big(4\log b/(1-b^{-2H})\big)^{1/2}$.

(ii) *For every $H \in (0,1)$ and every integer $b \geq 2$,*

$$M_H \geq \mu_b\,(2b)^{-H}\Big(\tfrac{1}{2}\coth\big(\tfrac{1}{2}H\log(2b)\big)\Big)^{1/2} \;\geq\; \frac{\mu_b\,(2b)^{-H}}{\sqrt{H\log(2b)}}. \tag{2.2}$$

(iii) *For every $H \in (0,\frac{1}{2}]$, $M_H \leq \sqrt{2/H} + \sqrt{H}(2+\log(1/H))$. For every $H \in (0, 3\cdot 10^{-6}]$, $M_H \geq \sqrt{2/H} - 2\sqrt{2}\sqrt{\log(1/H)}$. In particular*

$$\lim_{H\downarrow 0}\sqrt{H}\,M_H = \sqrt{2}.$$

(iv) *For all $H \in (0,1)$ and $u > 0$, $\mathbb{P}(|\sup_{[0,1]} B^H - M_H| > u) \leq 2e^{-u^2/2}$. Consequently $\sqrt{H}\sup_{[0,1]} B^H \to \sqrt{2}$ in L^p for every $p \in [1,\infty)$.*

The bounds (2.1)–(2.2) are elementary to evaluate; see Table 1. With $b = 2$ and $\delta = H$, (2.1) already gives an additive error $O(\sqrt{H}\log(1/H))$, which tends to zero. The lower bound (2.2) is optimised by letting $b \to \infty$ as $H \downarrow 0$.

Theorem 2.2 (Free energy and phase transition)**.** *Let $Z_H(\beta) := \int_0^1 e^{\beta B_t^H/\sqrt{H}}dt$ for $\beta \geq 0$. Then, for every $\beta \geq 0$ and $p \in [1,\infty)$,*

$$H\log Z_H(\beta) \longrightarrow F(\beta) := \begin{cases} \beta^2/4, & 0 \leq \beta \leq \sqrt{2}, \\ \sqrt{2}\,\beta - \frac{3}{2} - \log\big(\beta/\sqrt{2}\big), & \beta \geq \sqrt{2}, \end{cases} \qquad \text{in } L^p.$$

The function F is convex and C^2 on $(0,\infty)$, with $F'' = \frac{1}{2}$ on $(0,\sqrt{2})$ and $F''(\beta) = \beta^{-2}$ on $(\sqrt{2},\infty)$. Its third derivative jumps from 0 to $-1/\sqrt{2}$ at $\beta_c = \sqrt{2}$, and $F(\beta)/\beta \to \sqrt{2}$ as $\beta \to \infty$.

Remark 2.3 (Annealed free energy and the chaos parameter). Since $\mathbb{E}Z_H(\beta) = \int_0^1 e^{\beta^2 t^{2H}/(2H)}dt$, the annealed free energy is $H\log\mathbb{E}Z_H(\beta) \to \beta^2/2$, twice the quenched value $F(\beta) = \beta^2/4$ even at high temperature $\beta < \sqrt{2}$. The discrepancy is the contribution of the common mode of Section 1.2(a), which has variance close to $\frac{1}{2}$ and enters $\log Z_H(\beta)$ as an additive centred Gaussian term. The critical value $\sqrt{2}$ is best read off the covariance at mesoscopic scales. Expanding $|t-s|^{2H} = e^{-2H\log(1/|t-s|)}$ gives, for $H\log(1/|t-s|) \to 0$,

$$\frac{\mathrm{Cov}(B_s^H, B_t^H)}{H} = \frac{s^{2H}+t^{2H}-1}{2H} + \log\frac{1}{|t-s|} + O\Big(H\log^2\frac{1}{|t-s|}\Big).$$

Up to a common mode, $B^H/\sqrt{H}$ is therefore a log-correlated field with unit coefficient, β is the parameter usually called γ in Gaussian multiplicative chaos, and $\beta_c = \sqrt{2}$ is the critical value of chaos in dimension one [52, 69]. In the same correspondence H plays the role of $1/\log(1/\delta)$ for a log-correlated field with cutoff δ, since both count the effective scales. The general dictionary is given in Remark 2.8.

2.2. **A universality principle.** The mechanism of Section 1.2 is not specific to fBm. We formalise the class of variograms to which it applies.

Definition 2.4 (Profiles)**.** A *profile* is a function $\Phi \in C^1([0,\infty))$ which is convex, satisfies $\Phi'(y) < 0$ for all $y \geq 0$ and $\lim_{y\to\infty}\Phi(y) = 0$, and for which, writing $g := -\Phi'$,

$$I_\Phi := \int_0^\infty \sqrt{g(y)}\,dy < \infty.$$

For a profile, g is positive, continuous and non-increasing, and $\Phi(y) = \int_y^\infty g$. Since $\sqrt{g}$ is non-increasing and integrable, $y\sqrt{g(y)} \le 2\int_{y/2}^y \sqrt{g} \to 0$, hence $g(y) = o(y^{-2})$ and

$$\lim_{y\to\infty} y\,\Phi(y) = 0. \tag{2.3}$$

Convexity of Φ says that the variance per unit of log-scale, g, is non-increasing. It is the continuum counterpart of the *concave* profile condition for GREMs in Section 4: the cumulative variance $\frac{1}{2}(\Phi(0) - \Phi(y))$ is a concave function of the entropy, which is proportional to y.

Assumption 2.5. $(X^\varepsilon)_{0<\varepsilon<\varepsilon_0}$ is a family of centred Gaussian processes on $\mathsf{T} = [0,1]^d$ with continuous sample paths. There are a profile Φ, a function $\eta : (0,\varepsilon_0) \to [0,1)$ with $\eta(\varepsilon) \to 0$, a constant $C_0 < \infty$ and, for every ε, numbers $0 < q_\varepsilon \le r_\varepsilon \le 1$ and a closed cube $Q_\varepsilon \subset \mathsf{T}$ of side length q_ε, such that $\varepsilon\log(1/q_\varepsilon) \to 0$ as $\varepsilon \downarrow 0$ and, writing $\mathsf{d} = \mathsf{d}_{X^\varepsilon}$ and $\eta = \eta(\varepsilon)$,

$$\mathsf{d}(s,t)^2 \le \big(1+\eta\big)\,\Phi\big(\varepsilon\log(1/|t-s|)\big), \qquad s,t\in\mathsf{T},\ 0<|t-s|\le r_\varepsilon, \tag{2.4}$$

$$\mathsf{d}(s,t)^2 \ge \big(1-\eta\big)\,\Phi\big(\varepsilon\log(1/|t-s|)\big), \qquad s\ne t\in Q_\varepsilon, \tag{2.5}$$

$$\mathsf{d}(s,t)^2 \le C_0, \qquad s,t\in\mathsf{T}. \tag{2.6}$$

Remark 2.6.

(a) In most examples (2.4) holds for all $s \ne t$, i.e. with $r_\varepsilon = 1$, and (2.5) holds on a fixed cube. If $r_\varepsilon = 1$, then (2.6) follows from (2.4) with $C_0 = 2\Phi(0)$, since $|t-s| \le 1$ on T and Φ is decreasing.

(b) Since $q_\varepsilon \le r_\varepsilon$, also $\varepsilon\log(1/r_\varepsilon) \to 0$. In the log-scale variable $y = \varepsilon\log(1/|t-s|)$, the profile is therefore only required on $[\varepsilon\log(1/r_\varepsilon),\infty)$, and the lower bound only inside a cube that may shrink as $\varepsilon \downarrow 0$. A vanishing fraction of the effective scales, together with the behaviour of X^ε at macroscopic distances, does not affect the limits. For example, the mixed fractional Brownian motion $X^\varepsilon = B^\varepsilon + cW$ of Cheridito [27], with W a Brownian motion independent of B^ε and $c \ne 0$, violates (2.4) at distances of order one. For ε so small that $c^2 e^{-(1-2\varepsilon)/\sqrt{\varepsilon}} < 1$, it satisfies Assumption 2.5 with $\Phi(y) = e^{-2y}$, $q_\varepsilon = r_\varepsilon = e^{-1/\sqrt{\varepsilon}}$, $Q_\varepsilon = [0, q_\varepsilon]$, $\eta = c^2 r_\varepsilon^{1-2\varepsilon}$ and $C_0 = 1 + c^2$.

(c) *Perturbations.* If Y^ε are processes with continuous paths on T (not necessarily Gaussian, nor independent of X^ε) with $\mathbb{E}\sup_{\mathsf{T}}|Y^\varepsilon| = o(\varepsilon^{-1/2})$, then the conclusions (i) and (ii) of Theorem 2.7 below transfer from X^ε to $X^\varepsilon + Y^\varepsilon$, because $|\sup(X+Y) - \sup X| \le \sup|Y|$ and $|\log\int e^{\lambda(X+Y)} - \log\int e^{\lambda X}| \le \lambda\sup|Y|$.

(d) *Brackets.* The upper bounds in the proof of Theorem 2.7 use only (2.4) and (2.6), and the lower bounds use (2.5), together with (2.4) at scales below the size of the cells. Hence, if (2.5) holds with a profile Φ_1 and (2.4) with a profile Φ_2, then the limits inferior in Theorem 2.7 are at least the values for Φ_1, and the limits superior are at most the values for Φ_2.

For $d \ge 1$ define the one-level free energy function

$$G_d(z) := \begin{cases} z^2/4, & 0 \le z \le 2\sqrt{d}, \\ \sqrt{d}\,z - d, & z \ge 2\sqrt{d}, \end{cases} \qquad \text{and} \qquad F_{\Phi,d}(\beta) := \int_0^\infty G_d\big(\beta\sqrt{g(y)}\big)\,dy. \tag{2.7}$$

G_d is convex, C^1, non-decreasing, and satisfies $G_d(z) \le \min(z^2/4, \sqrt{d}\,z)$. Hence $0 \le F_{\Phi,d}(\beta) \le \min(\beta^2\Phi(0)/4,\ \beta\sqrt{d}\,I_\Phi)$.

Theorem 2.7 (Universality)**.** *Under Assumption 2.5:*

(i) $\lim_{\varepsilon\downarrow 0} \sqrt{\varepsilon}\,\mathbb{E}\sup_{t\in\mathsf{T}} X_t^\varepsilon = \sqrt{d}\,I_\Phi$.

(ii) *For every* $\beta \geq 0$, $\lim_{\varepsilon \downarrow 0} \varepsilon\, \mathbb{E} \log \int_{\mathsf{T}} \exp\bigl(\beta X_t^\varepsilon / \sqrt{\varepsilon}\bigr) dt = F_{\Phi,d}(\beta)$.

(iii) *If moreover* $\sup_{t \in \mathsf{T}} \operatorname{Var}(X_t^\varepsilon) = o(1/\varepsilon)$, *the convergences in* (i) *and* (ii) *hold in* L^p *for the random variables* $\sqrt{\varepsilon} \sup_{\mathsf{T}} X^\varepsilon$ *and* $\varepsilon \log \int_{\mathsf{T}} e^{\beta X^\varepsilon/\sqrt{\varepsilon}}$, *for every* $p \in [1, \infty)$.

Remark 2.8 (The continuous random energy model). The limits in Theorem 2.7 are those of the continuous random energy model (CREM) of Bovier and Kurkova [21]. There, the index set is $\{-1, 1\}^N$, the entropy is $N \log 2$ per unit of the overlap variable $x \in [0, 1]$, and the covariance of two configurations is $N\, A(x)$ for a non-decreasing function A, where x is the normalised depth of their deepest common ancestor. In our setting the overlap variable is the log-scale $y = \varepsilon \log(1/|t - s|) \in [0, \infty)$, which carries entropy d/ε per unit, and the role of A is played by the cumulative variance

$$A(y) := \tfrac{1}{2}\bigl(\Phi(0) - \Phi(y)\bigr), \qquad \text{so that} \qquad \mathsf{d}(s,t)^2 \approx 2\bigl(A(\infty) - A(y)\bigr).$$

Convexity of Φ is concavity of A, the case in which the formulas of [21] involve no concave hull. With this dictionary, $\sqrt{d}\, I_\Phi$ is the ground-state energy of [21, Thm 3.1], and $F_{\Phi,d}$ and β_c are the free energy and the critical inverse temperature of [21, Thm 3.3]; our free energy is normalised by the volume of T rather than by the number of configurations. What Theorem 2.7 adds is that these formulas hold for non-hierarchical processes on $[0, 1]^d$ that are specified only through variogram bounds, in the regime of a diverging number of scales of vanishing variance.

As in Remark 2.3, if $\mathsf{d}(s,t)^2 \approx \Phi(\varepsilon \log(1/|t-s|))$, then $\operatorname{Cov}(X_s^\varepsilon, X_t^\varepsilon)/\varepsilon$ equals $\frac{1}{2} g(0) \log(1/|t-s|)$ up to a common mode and lower-order terms. So β corresponds to the chaos parameter $\gamma = \beta\sqrt{g(0)/2}$, and $\beta_c = 2\sqrt{d}/\sqrt{g(0)}$ in Proposition 2.9 below corresponds to the critical value $\gamma_c = \sqrt{2d}$ of Gaussian multiplicative chaos in dimension d [69].

Proposition 2.9 (Structure of the free energy). *Let* Φ *be a profile and* $\beta_c := 2\sqrt{d}/\sqrt{g(0)}$.

(a) $F := F_{\Phi,d}$ *is convex, non-decreasing and* C^1 *on* $[0, \infty)$, *with*

$$F'(\beta) = \int_0^\infty \min\bigl(\beta g(y)/2, \sqrt{d\, g(y)}\bigr)\, dy.$$

Moreover $F(\beta) = \Phi(0)\beta^2/4$ *for* $\beta \leq \beta_c$, *and* $F(\beta)/\beta \to \sqrt{d}\, I_\Phi$ *as* $\beta \to \infty$.

(b) *If* Φ *is strictly convex, then* $F \in C^2((0, \infty))$ *and, for* $\beta > \beta_c$, $F''(\beta) = \frac{1}{2}\Phi(y_*(\beta))$, *where* $y_*(\beta) > 0$ *is the unique solution of* $\beta\sqrt{g(y)} = 2\sqrt{d}$. *For* $\beta < \beta_c$, $F''(\beta) = \frac{1}{2}\Phi(0)$.

(c) *If moreover* $\Phi \in C^2([0, y_0))$ *for some* $y_0 > 0$ *and* $\Phi''(0) > 0$, *then* F''' *exists on* $(0, \beta_c) \cup (\beta_c, \beta_c + \theta)$ *for some* $\theta > 0$, *vanishes on* $(0, \beta_c)$, *and* $\lim_{\beta \downarrow \beta_c} F'''(\beta) = -\Phi'(0)^2/(\beta_c \Phi''(0)) < 0$. *Thus* F *undergoes a third-order phase transition at* β_c.

Here the order of a phase transition is the order of the lowest derivative of $\beta \mapsto F(\beta)$ that is discontinuous. In this convention the random energy model, whose free energy is $\beta \mapsto \mathsf{R}(\beta\sqrt{\ell}; \ell)/\ell$ with R as in (3.4) below, has a transition of second order. Part (a) records the high-temperature branch of the CREM free energy [21, Thm 3.3]. Parts (b) and (c) are elementary consequences of the formula; we have not found them stated explicitly in the literature.

Remark 2.10 (The order of the transition). The behaviour of g near 0 determines the order of the transition.

(a) If $\Phi''(0) > 0$, the transition is of third order, by Proposition 2.9(c).

(b) If Φ is strictly convex and C^3 near 0 with $\Phi''(0) = 0$ and $\Phi'''(0) > 0$, then $g(0) - g(y) \sim \frac{1}{2}\Phi'''(0)y^2$ as $y \downarrow 0$, and Proposition 2.9(b) gives

$$F''(\beta_c) - F''(\beta) \sim g(0)^{3/2}\Big(\frac{\beta - \beta_c}{\beta_c\,\Phi'''(0)}\Big)^{1/2} \qquad (\beta \downarrow \beta_c).$$

So F is C^2, but $F'''(\beta) \to -\infty$ as $\beta \downarrow \beta_c$. This is the case for the powered-exponential fields of Corollary 2.11(d).

(c) The linear profile $\Phi(y) = 2(1-y)^+$ is not a profile in the sense of Definition 2.4: it is not C^1 at $y = 1$, and $\Phi' = 0$ on $(1, \infty)$. It is the rescaled variogram of a log-correlated field Y with cutoff δ: if $\mathsf{d}_Y(s,t)^2 = 2\log(|t-s|/\delta)$ for $|t-s| \geq \delta$ and $\varepsilon := 1/\log(1/\delta)$, then $\varepsilon\,\mathsf{d}_Y(s,t)^2 = \Phi(\varepsilon\log(1/|t-s|))$ for $|t-s| \geq \delta$. Below the cutoff, (2.4) fails, since $\Phi(\varepsilon\log(1/|t-s|)) = 0$ while $\mathsf{d}_Y(s,t) > 0$, so Theorem 2.7 does not apply even with the relaxed hypotheses. Formally, the formulas of Theorem 2.7 give $\sqrt{d}\,I_\Phi = \sqrt{2d}$ and $F_{\Phi,d}(\beta) = G_d(\sqrt{2}\beta)$. These are the first-order maximum of log-correlated fields [56, 39] and their freezing free energy [36, 24, 46, 57], whose transition at $\beta_c = \sqrt{2d}$ is of second order.

Curvature of the profile at the coarsest scale thus smooths the second-order freezing transition of log-correlated fields. Consistently, the limit of F''' in Proposition 2.9(c) diverges as $\Phi''(0) \downarrow 0$.

Corollary 2.11 (Examples)**.**

(a) *Fractional Brownian fields. Let X^H be the Lévy fractional Brownian field on $[0,1]^d$, i.e. the centred Gaussian field with $X_0^H = 0$ and $\mathbb{E}|X_t^H - X_s^H|^2 = \|t-s\|_2^{2H}$. Then $\sqrt{H}\,\mathbb{E}\sup_{[0,1]^d} X^H \to \sqrt{2d}$, and $H\log\int_{[0,1]^d} e^{\beta X_t^H/\sqrt{H}}dt \to F_d(\beta)$ in L^p, where $F_d(\beta) = \beta^2/4$ for $\beta \leq \sqrt{2d}$ and $F_d(\beta) = \sqrt{2d}\,\beta - \frac{3}{2}d - d\log(\beta/\sqrt{2d})$ for $\beta \geq \sqrt{2d}$.*

(b) *Riemann–Liouville processes. Let $R_t^H = \sqrt{2H}\int_0^t (t-u)^{H-1/2}dW_u$, $t \in [0,1]$, for a Brownian motion W. Then $\sqrt{H}\,\mathbb{E}\sup_{[0,1]} R^H \to 2$, and $H\log\int_0^1 e^{\beta R_t^H/\sqrt{H}}dt \to F_{RL}(\beta)$ in L^p, with $F_{RL}(\beta) = \beta^2/2$ for $\beta \leq 1$ and $F_{RL}(\beta) = 2\beta - \frac{3}{2} - \log\beta$ for $\beta \geq 1$.*

(c) *Mixtures of fractional Brownian motions. Let μ be a non-zero finite Borel measure on a compact interval $[h_-, h_+] \subset (0,\infty)$. For $\varepsilon < 1/h_+$ let X^ε be a centred Gaussian process on $[0,1]$ with $X_0^\varepsilon = 0$, stationary increments and $\mathbb{E}(X_t^\varepsilon - X_s^\varepsilon)^2 = \int |t-s|^{2\varepsilon h}\mu(dh)$. (Its covariance is a mixture of fBm covariances, so X^ε exists.) Then, with $\Phi_\mu(y) := \int e^{-2hy}\mu(dh)$,*

$$\sqrt{\varepsilon}\,\mathbb{E}\sup_{[0,1]} X^\varepsilon \longrightarrow \int_0^\infty \Big(\int 2h\,e^{-2hy}\mu(dh)\Big)^{1/2}dy,$$

and $\varepsilon\log\int_0^1 e^{\beta X_t^\varepsilon/\sqrt{\varepsilon}}dt \to F_{\Phi_\mu,1}(\beta)$ in L^p. For $\mu = \sum_k c_k^2\delta_{h_k}$ the process X^ε is the superposition $\sum_k c_k B^{(k)}$ of independent fBms with Hurst indices εh_k.

(d) *Powered-exponential fields. For $H \in (0,1]$ let X^H be the centred stationary Gaussian field on $[0,1]^d$ with $\mathrm{Cov}(X_s^H, X_t^H) = \exp(-\|t-s\|_2^{2H})$. Then*

$$\sqrt{H}\,\mathbb{E}\sup_{[0,1]^d} X^H \longrightarrow 2\sqrt{d}\int_0^1 e^{-u^2/2}\,du \approx 1.7112\,\sqrt{d},$$

and $H\log\int_{[0,1]^d} e^{\beta X_t^H/\sqrt{H}}dt \to F_{\Phi,d}(\beta)$ in L^p, where $\Phi(y) = 2(1-\exp(-e^{-2y}))$ and $\beta_c = \sqrt{e\,d}$. Here $\Phi''(0) = 0$, and the transition at β_c is of the type described in Remark 2.10(b).

The Riemann–Liouville constant 2 exceeds the fBm constant $\sqrt{2}$ because R^H has no common mode: $\mathrm{Cov}(R_s^H, R_t^H) \to 0$ for $s \neq t$ as $H \downarrow 0$. For the powered-exponential field the common mode has variance $1 - \frac{1}{2}\Phi(0) = e^{-1}$, and its profile Φ is not a positive mixture of exponentials,

so this example is not a mixture of fBms. The normalisation of R^H gives $\operatorname{Var} R_1^H = 1$. In the normalisation of fractional calculus, $\widetilde{R}_t^H = \Gamma(H+\frac{1}{2})^{-1}\int_0^t (t-u)^{H-1/2}dW_u = R_t^H/(\sqrt{2H}\,\Gamma(H+\frac{1}{2}))$, Corollary 2.11(b) reads $H\,\mathbb{E}\sup_{[0,1]}\widetilde{R}^H \to \sqrt{2/\pi}$; note the different power of H. In the chaos parametrisation of Remark 2.8, R^H has $g(0)=4$, so $\gamma = \sqrt{2}\,\beta$, and $\beta_c = 1$ corresponds to the critical value $\gamma_c = \sqrt{2}$ of one-dimensional chaos.

2.3. **Maxima over grids.** For $n \geq 1$ write $\Gamma_n := \{i/n : 1 \leq i \leq n\}$, and define

$$\Lambda(\kappa) := \begin{cases} \sqrt{\kappa}, & 0 \leq \kappa \leq \frac{1}{2}, \\ \sqrt{2} - \frac{1}{\sqrt{2}} e^{1/2-\kappa}, & \kappa \geq \frac{1}{2}, \end{cases} \qquad \Lambda(\infty) := \sqrt{2}. \tag{2.8}$$

Proposition 2.12. *For every $H \in (0,1)$ and every set $\Gamma \subset [0,1]$ of n points, $\mathbb{E}\max_{t\in\Gamma} B_t^H \leq \mu_n/\sqrt{2} \leq \sqrt{\log n}$.*

Theorem 2.13 (Discretisation phase diagram)**.** *Let $H \downarrow 0$ and $n = n(H) \to \infty$ with $H\log n \to \kappa \in [0,\infty]$. Then*

$$\sqrt{H}\;\mathbb{E}\max_{t\in\Gamma_n} B_t^H \longrightarrow \Lambda(\kappa).$$

The function Λ is concave, increasing, continuous on $[0,\infty)$ and C^2 on $(0,\infty)$, with $\Lambda'(\frac{1}{2}) = 1/\sqrt{2}$ and $\Lambda''(\frac{1}{2}) = -1/\sqrt{2}$. Its third derivative jumps from $3/\sqrt{2}$ to $1/\sqrt{2}$ at $\kappa = \frac{1}{2}$. For $\kappa \leq \frac{1}{2}$ the grid maximum behaves like the maximum of n independent $N(0,\frac{1}{2})$ variables. This extends the regime $H(\log n)^2 \to 0$, in which Borovkov and Zhitlukhin [15] proved a normal approximation for the grid maximum. For $\kappa > \frac{1}{2}$ the coarse scales $y \leq \kappa - \frac{1}{2}$ are resolved hierarchically, and the remaining scales are seen as a single random energy model. Since $\Lambda(\kappa) < \sqrt{2}$ for every $\kappa < \infty$, capturing M_H to first order requires $\log n \gg 1/H$ grid points. This explains, and quantifies, the "simulation paradoxes" of [59]. In terms of the discretisation error, Theorems 2.1 and 2.13 give the following.

Corollary 2.14 (Discretisation bias)**.** *Let $H \downarrow 0$ and $H\log n \to \kappa \in [0,\infty)$. Then*

$$\sqrt{H}\left(M_H - \mathbb{E}\max_{t\in\Gamma_n} B_t^H\right) \longrightarrow \sqrt{2} - \Lambda(\kappa) > 0.$$

In particular, if $\kappa \geq \frac{1}{2}$, then $M_H - \mathbb{E}\max_{\Gamma_n} B^H = (1+o(1))\sqrt{e/(2H)}\;n^{-H}$.

Proof. The first claim combines Theorem 2.1(iii) and Theorem 2.13. If $\kappa \geq \frac{1}{2}$, then $\sqrt{2}-\Lambda(\kappa) = e^{1/2-\kappa}/\sqrt{2}$, which is the limit of $\sqrt{H}\cdot\sqrt{e/(2H)}\,n^{-H} = \sqrt{e/2}\,e^{-H\log n}$; since the limit is positive, the ratio tends to one. □

Remark 2.15 (Deterministic designs). For fixed $H \in (0,\frac{1}{2})$, Borovkov, Mishura, Novikov and Zhitlukhin [14, Thm 1 and Rem. 3] showed that $M_H - \mathbb{E}\max_{\Gamma_n} B^H \leq (1+o(1))\,n^{-H}\sqrt{\log n}$ as $n \to \infty$. Corollary 2.14 shows that this asymptotic form is not uniform in H: if $H\log n \to \kappa \in [\frac{1}{2}, \frac{e}{2})$, then $\sqrt{H}\,n^{-H}\sqrt{\log n} \to \sqrt{\kappa}\,e^{-\kappa}$, which is smaller than the true limit $e^{1/2-\kappa}/\sqrt{2}$. For $\kappa > \frac{e}{2}$ the order is reversed, and the true limit is smaller than $\sqrt{\kappa}\,e^{-\kappa}$ by the factor $\sqrt{e/(2\kappa)}$. Proposition 2.12 also gives a statement that does not depend on the design. If $\Gamma \subset [0,1]$ is a deterministic set of n points with $\mathbb{E}\max_\Gamma B^H \geq \theta\sqrt{2/H}$ for some $\theta \in (0,1)$, then $\log n \geq 2\theta^2/H$. For $\theta \leq \frac{1}{2}$ uniform grids attain this threshold to first order, since $\Lambda(2\theta^2) = \sqrt{2}\,\theta$. For $\theta \in (\frac{1}{2},1)$ uniform grids need $H\log n \to \frac{1}{2} + \log\frac{1}{2(1-\theta)}$, and we do not know whether other deterministic designs can do better. Designs that depend on the sampled path are not covered by Proposition 2.12.

2.4. **Pickands' constants.** For $\alpha \in (0,2]$ let $\mathcal{H}_\alpha$ be Pickands' constant [68]. Shao [72, Lemma 4] proved that $\mathcal{H}_\alpha \le \big(\frac{\alpha e}{\sqrt{2}}\,\mathbb{E}\sup_{[0,1]} B^{\alpha/2}\big)^{2/\alpha}$ for $\alpha \in (0,2)$. With $\alpha = 2H$ this reads $\mathcal{H}_{2H} \le (\sqrt{2}\,e\,H\,M_H)^{1/H}$; see also [14, Rem. 6]. Combined with Theorem 2.1(iii) this gives:

Corollary 2.16. *For $H \in (0, \frac{1}{2}]$, $\mathcal{H}_{2H} \le (2e\sqrt{H})^{1/H} \exp\big((2+\log(1/H))/\sqrt{2}\big)$. In particular*

$$\limsup_{H\downarrow 0}\Big(H\log\mathcal{H}_{2H} - \tfrac{1}{2}\log H\Big) \le \log(2e) \approx 1.693.$$

For comparison, Shao's explicit bound [72, (1.5)] gives the value ≈ 2.31 for this limit superior, and the bound $\mathcal{H}_{2H} < (42.46H)^{1/(2H)}$ of [14, Rem. 6] gives $\frac{1}{2}\log 42.46 \approx 1.874$. Harper's lower bound $\mathcal{H}_\alpha \ge 1.1527^{1/\alpha}/\Gamma(1/\alpha)$, valid for all sufficiently small α [51], gives $\liminf(H\log\mathcal{H}_{2H} - \frac{1}{2}\log H) \ge \frac{1}{2}\log(2e\cdot 1.1527) \approx 0.918$.

3. Preliminaries

3.1. **Comparison of free energies.** The following comparison is the basic tool of the paper. It is the interpolation identity in the proof of [25, Thm 1.2] (see (1) there), which allows arbitrary variances and equal, possibly non-zero, means; the weights w_i can be absorbed into the means. Comparison inequalities of this type for free energies go back to Kahane [52, 53], and the interpolation method is a standard tool for spin glasses [48, 74]. We include the short proof for completeness, since the version for *free energies with arbitrary variances* is essential for us.

Lemma 3.1. *Let $X = (X_i)_{i\le n}$ and $Y = (Y_i)_{i\le n}$ be centred Gaussian vectors, $w \in (0,\infty)^n$ and $\lambda \in \mathbb{R}$. If $\mathbb{E}(X_i - X_j)^2 \le \mathbb{E}(Y_i - Y_j)^2$ for all i, j, then*

$$\mathbb{E}\log\sum_{i=1}^n w_i e^{\lambda X_i} \;\le\; \mathbb{E}\log\sum_{i=1}^n w_i e^{\lambda Y_i}.$$

Proof. Take X and Y independent and set $Z(s) = \sqrt{1-s}\,X + \sqrt{s}\,Y$ for $s \in [0,1]$, $F(z) = \log\sum_i w_i e^{\lambda z_i}$ and $\varphi(s) = \mathbb{E}F(Z(s))$. Write $\pi_i(z) = w_i e^{\lambda z_i}/\sum_j w_j e^{\lambda z_j}$, so that $\partial_i F = \lambda\pi_i$ and $\partial_{ij}F = \lambda^2(\delta_{ij}\pi_i - \pi_i\pi_j)$; in particular $|\partial_i F| \le |\lambda|$ and $|\partial_{ij}F| \le \lambda^2$. Since $|F(z)| \le |\lambda|\max_i|z_i| + \max_i|\log w_i| + \log n$, dominated convergence shows that φ is continuous on $[0,1]$. Let $\dot Z_i(s) = Y_i/(2\sqrt{s}) - X_i/(2\sqrt{1-s})$. Since $|\frac{d}{ds}F(Z(s))| \le |\lambda|\sum_i|\dot Z_i(s)|$, which is dominated uniformly for s in compact subsets of $(0,1)$, φ is differentiable on $(0,1)$ with $\varphi'(s) = \sum_i \mathbb{E}[\partial_i F(Z(s))\dot Z_i(s)]$. By Gaussian integration by parts, applied to the centred Gaussian vector $(\dot Z(s), Z(s))$,

$$\mathbb{E}[\dot Z_i(s)\,\partial_i F(Z(s))] = \sum_j \mathbb{E}[\dot Z_i(s) Z_j(s)]\,\mathbb{E}[\partial_{ij}F(Z(s))], \qquad \mathbb{E}[\dot Z_i Z_j] = \tfrac{1}{2}(C^Y_{ij} - C^X_{ij}),$$

where C^X, C^Y are the covariance matrices. Write $\Delta C = C^Y - C^X$ and $\Delta D_{ij} = \mathbb{E}(Y_i - Y_j)^2 - \mathbb{E}(X_i - X_j)^2 = \Delta C_{ii} + \Delta C_{jj} - 2\Delta C_{ij}$. Then

$$\varphi'(s) = \frac{\lambda^2}{2}\,\mathbb{E}\Big[\sum_i \Delta C_{ii}\pi_i - \sum_{i,j}\Delta C_{ij}\pi_i\pi_j\Big].$$

Using $\sum_j \pi_j = 1$ we have $\sum_{i,j}\Delta C_{ij}\pi_i\pi_j = \sum_i \Delta C_{ii}\pi_i - \frac{1}{2}\sum_{i,j}\Delta D_{ij}\pi_i\pi_j$. Hence

$$\varphi'(s) = \frac{\lambda^2}{4}\sum_{i,j}\Delta D_{ij}\,\mathbb{E}\big[\pi_i(Z(s))\pi_j(Z(s))\big] \;\ge 0, \tag{3.1}$$

and $\varphi(0) \le \varphi(1)$. □

Corollary 3.2 (Sudakov–Fernique)**.** *Let X, Y be centred Gaussian processes indexed by a countable set D, with $\mathbb{E}(X_s - X_t)^2 \le \mathbb{E}(Y_s - Y_t)^2$ for $s, t \in D$ and $\mathbb{E}\sup_D X < \infty$. Then $\mathbb{E}\sup_D X \le \mathbb{E}\sup_D Y$.*

Proof. For finite D, divide the inequality of Lemma 3.1 (with $w \equiv 1$) by λ and let $\lambda \to \infty$. Since $\max_i z_i \le \lambda^{-1}\log\sum_i e^{\lambda z_i} \le \max_i z_i + \lambda^{-1}\log n$, we get $\mathbb{E}\max X \le \lambda^{-1}\mathbb{E}\log\sum_i e^{\lambda X_i} \le \lambda^{-1}\mathbb{E}\log\sum_i e^{\lambda Y_i} \le \mathbb{E}\max Y + \lambda^{-1}\log n$ for every $\lambda > 0$, hence $\mathbb{E}\max X \le \mathbb{E}\max Y$. For countable D, take finite $D_k \uparrow D$ and use monotone convergence, noting that $\max_{D_k} X \ge X_{t_0}$ for a fixed $t_0 \in D_1$. □

3.2. **Concentration.** Parts (a) and (b) of the following lemma are classical [12, 76]; see also [1, Thm. 2.1.1]. Part (c) follows from (a), together with the integrability statement in [1, Thm. 2.1.1].

Lemma 3.3.

(a) *Let ξ be a standard Gaussian vector in $\mathbb{R}^m$ and $f : \mathbb{R}^m \to \mathbb{R}$ be L-Lipschitz. Then $\mathbb{P}(f(\xi) - \mathbb{E}f(\xi) \ge u) \le e^{-u^2/(2L^2)}$ and $\mathbb{P}(f(\xi) - \mathbb{E}f(\xi) \le -u) \le e^{-u^2/(2L^2)}$ for $u > 0$.*

(b) (Borell–TIS) *If X is a centred Gaussian process on a countable set with $\mathbb{E}\sup X < \infty$ and $\sigma^2 := \sup_t \operatorname{Var} X_t < \infty$, then $\mathbb{P}(|\sup X - \mathbb{E}\sup X| > u) \le 2e^{-u^2/(2\sigma^2)}$.*

(c) *Let X be a centred Gaussian process on T with continuous paths and $\sigma^2 := \sup_t \operatorname{Var} X_t$, and let $\lambda \ge 0$. Then $f := \log\int_{\mathsf{T}} e^{\lambda X_t} dt$ satisfies $\mathbb{P}(|f - \mathbb{E}f| > u) \le 2e^{-u^2/(2\lambda^2\sigma^2)}$ for $u > 0$.*

Proof of (c). For a fixed integer $b \ge 2$ let $\mathcal{Q}_K$ be the partition of T into b^{dK} half-open b-adic cubes, as in Section 4, with lower-left corners t_Q. The Riemann sums $f_K := \log\sum_{Q\in\mathcal{Q}_K} b^{-dK} e^{\lambda X_{t_Q}}$ are Lipschitz functions of a standard Gaussian vector with constant $\lambda\sigma$ (see below), so (a) applies to them with constants independent of K. By path continuity $f_K \to f$ almost surely, and $|f_K| \le \lambda \sup_{\mathsf{T}} |X|$. Since X has continuous paths, $\sup_{\mathsf{T}} |X| = \sup_D |X|$ for a countable dense set $D \subset \mathsf{T}$, and this is almost surely finite, so Borell–TIS on D [1, Thm 2.1.1] gives $\mathbb{E}\sup_{\mathsf{T}} |X| < \infty$. So $\mathbb{E}f_K \to \mathbb{E}f$ by dominated convergence, and Fatou's lemma gives $\mathbb{P}(|f - \mathbb{E}f| > u) \le \liminf_K \mathbb{P}(|f_K - \mathbb{E}f_K| > u) \le 2e^{-u^2/(2\lambda^2\sigma^2)}$. □

If $X = A\xi$ is a Gaussian vector with rows A_i, then $\xi \mapsto \max_i (A\xi)_i$ and $\xi \mapsto \log\sum_i w_i e^{\lambda(A\xi)_i}$ are Lipschitz with constants $\max_i \|A_i\| = \max_i \sigma_i$ and $|\lambda| \max_i \sigma_i$ respectively: the gradient of the latter is $\lambda\sum_i \pi_i A_i$, a convex combination of the vectors λA_i. We record two elementary consequences of Lemma 3.3(a): for an L-Lipschitz f and $p \in (0,1)$,

$$\mathbb{P}(f(\xi) \le a) > \tfrac{1}{2} \Rightarrow \mathbb{E}f(\xi) \le a + L\sqrt{2\log 2}, \quad \mathbb{P}(f(\xi) \ge a) > p \Rightarrow \mathbb{E}f(\xi) \ge a - L\sqrt{2\log(1/p)}. \tag{3.2}$$

For instance, if $\mathbb{E}f > a + L\sqrt{2\log 2}$ then $\mathbb{P}(f \le a) \le \mathbb{P}(f - \mathbb{E}f \le -L\sqrt{2\log 2}) \le \frac{1}{2}$. We also use the following maximal inequality: if $Z_1, \dots, Z_N$ are integrable random variables with $\mathbb{P}(|Z_i - \mathbb{E}Z_i| > u) \le 2e^{-u^2/(2s^2)}$ for all i and $u > 0$, then

$$\mathbb{E}\max_{i\le N} Z_i \le \max_{i\le N} \mathbb{E}Z_i + s\Big(1 + \sqrt{2\log(2N)}\Big). \tag{3.3}$$

Indeed, with $u_0 := s\sqrt{2\log(2N)} \ge s$ we have

$$\mathbb{E}\max_i (Z_i - \mathbb{E}Z_i)^+ \le \int_0^\infty \min\big(1, 2Ne^{-u^2/(2s^2)}\big) du \le u_0 + 2N\int_{u_0}^\infty e^{-u^2/(2s^2)} du \le u_0 + \frac{s^2}{u_0}.$$

3.3. **Maxima of independent Gaussians.**

Lemma 3.4.

(a) *If $Z_1, \dots, Z_N$ satisfy $\mathbb{E}e^{\lambda Z_i} \le e^{\lambda^2\sigma^2/2}$ for all $\lambda \ge 0$, then $\mathbb{E}\max_i Z_i \le \sigma\sqrt{2\log N}$. In particular $\mu_n \le \sqrt{2\log n}$.*

(b) *If $\ell := \log n \ge 1000$, then $\mu_n \ge \sqrt{2\ell} - (\log\ell + 3)/\sqrt{2\ell}$.*

(c) *$n \mapsto \mu_n$ is non-decreasing and $\mu_n/\sqrt{2\log n} \to 1$.*

Proof. (a) For $\lambda > 0$, Jensen gives $\mathbb{E}\max_i Z_i \le \lambda^{-1}\log\sum_i \mathbb{E}e^{\lambda Z_i} \le \lambda^{-1}\log N + \lambda\sigma^2/2$. Take $\lambda = \sigma^{-1}\sqrt{2\log N}$ if $N \ge 2$ (for $N = 1$ let $\lambda \downarrow 0$).

(b) Let $x = \sqrt{2\ell} - (\log\ell + 2)/\sqrt{2\ell}$, so that $x \ge \sqrt{2\ell} - 1 \ge 43$ and $x^2 \le 2\ell - 2\log\ell - 4 + (\log\ell+2)^2/(2\ell) \le 2\ell - 2\log\ell - 3.95$; here we used that $(\log\ell+2)/\sqrt{2\ell} \le 1$ and that $(\log\ell+2)^2/(2\ell)$ is decreasing in $\ell \ge 1$ and at most 0.04 for $\ell \ge 1000$. The Mills-ratio bound $\mathbb{P}(\xi > x) \ge \frac{x}{1+x^2}\phi(x) \ge \phi(x)/(1.03\sqrt{2\ell})$, valid since $x \le \sqrt{2\ell}$ and $1/x \le 0.03\sqrt{2\ell}$, gives

$$n\,\mathbb{P}(\xi > x) \;\ge\; \frac{n}{1.03\sqrt{2\ell}\sqrt{2\pi}}\,e^{-\ell+\log\ell+1.975} \;=\; \frac{e^{1.975}}{2.06\sqrt{\pi}}\sqrt{\ell} \;\ge\; \sqrt{\ell}.$$

Hence $q := \mathbb{P}(\max_{i\le n}\xi_i \le x) = (1-\mathbb{P}(\xi > x))^n \le e^{-\sqrt{\ell}}$. Since $\max_i \xi_i \ge \xi_1$,

$$\mu_n \;\ge\; x\,\mathbb{P}(\max_i \xi_i > x) + \mathbb{E}[\xi_1 \mathbf{1}\{\max_i \xi_i \le x\}] \;\ge\; x(1-q) - \sqrt{q} \;\ge\; x - (\sqrt{2\ell}+1)e^{-\sqrt{\ell}/2},$$

by Cauchy–Schwarz. Finally $(\sqrt{2\ell}+1)e^{-\sqrt{\ell}/2} \le 1/\sqrt{2\ell}$ for $\ell \ge 1000$, because $s \mapsto (2s^2 + \sqrt{2}s)e^{-s/2}$ is decreasing for $s \ge 8$ and is less than 10^{-3} at $s = \sqrt{1000}$.

(c) Monotonicity is clear, and the limit follows from (a) and (b). □

3.4. **A one-level random energy estimate.** For $\ell > 0$ and $\lambda \ge 0$ let

$$\mathsf{R}(\lambda;\ell) := \begin{cases} \lambda^2/2, & \lambda \le \sqrt{2\ell}, \\ \lambda\sqrt{2\ell} - \ell, & \lambda \ge \sqrt{2\ell}. \end{cases} \tag{3.4}$$

$\mathsf{R}(\cdot;\ell)$ is convex and C^1, and the line $\lambda \mapsto \lambda\sqrt{2\ell} - \ell$ is its tangent at $\sqrt{2\ell}$. Consequently $\mathsf{R}(\lambda;\ell) \ge \lambda\sqrt{2\ell} - \ell$ for all λ. Moreover $\mathsf{R}(\cdot;\ell)$ is non-decreasing on $[0,\infty)$; in particular $\alpha \mapsto \mathsf{R}(\lambda\sqrt{\alpha};\ell)$ is non-decreasing for every $\lambda \ge 0$. The function $\beta \mapsto \mathsf{R}(\beta\sqrt{\ell};\ell)/\ell$ is the limiting quenched free energy of Derrida's random energy model with $n = e^\ell$ configurations [32]. We need a non-asymptotic lower bound.

Lemma 3.5. *Let $n \ge 2$, $\ell = \log n$, and $\psi_n(\lambda) := \mathbb{E}\log\big(\frac{1}{n}\sum_{i=1}^n e^{\lambda\xi_i}\big)$ with ξ_i i.i.d. $N(0,1)$. For every $\lambda \ge 0$ and every $\upsilon \in (0, \frac12]$ with $e^{-\upsilon^2\ell} \le \frac14$,*

$$\mathsf{R}(\lambda;\ell) - \psi_n(\lambda) \;\le\; \max\Big\{40(1+\ell)e^{-\upsilon^2\ell},\; \upsilon^2\ell + \lambda\big(\sqrt{2\ell} - \mu_n\big)\Big\}.$$

Consequently, for every $\kappa_0 < \infty$, $\varrho_n(\kappa_0) := \sup_{0\le\lambda\le\kappa_0\sqrt{\ell}} \dfrac{\mathsf{R}(\lambda;\ell) - \psi_n(\lambda)}{\ell} \longrightarrow 0$ as $n \to \infty$.

Proof. Case $\lambda \ge (1-\upsilon)\sqrt{2\ell}$. Keeping only the largest term, $\psi_n(\lambda) \ge \lambda\mu_n - \ell$. If $\lambda \ge \sqrt{2\ell}$, then $\mathsf{R} - (\lambda\mu_n - \ell) = \lambda(\sqrt{2\ell} - \mu_n)$. If $(1-\upsilon)\sqrt{2\ell} \le \lambda < \sqrt{2\ell}$, then $\mathsf{R} - (\lambda\mu_n - \ell) = \frac12(\lambda - \sqrt{2\ell})^2 + \lambda(\sqrt{2\ell} - \mu_n) \le \upsilon^2\ell + \lambda(\sqrt{2\ell} - \mu_n)$.

Case $\lambda < (1-\upsilon)\sqrt{2\ell}$. Here $\mathsf{R} = \lambda^2/2$. Let $u = \sqrt{2\ell}$,

$$\widehat{W} := \frac1n\sum_i e^{\lambda\xi_i - \lambda^2/2}, \qquad W := \frac1n\sum_i e^{\lambda\xi_i - \lambda^2/2}\mathbf{1}\{\xi_i \le u\} \le \widehat{W},$$

so that $\psi_n(\lambda) = \lambda^2/2 + \mathbb{E}\log\widehat{W}$. We have $p := \mathbb{E}W = \mathbb{P}(\xi \le u-\lambda) \ge 1 - e^{-(u-\lambda)^2/2} \ge 1 - e^{-\upsilon^2\ell} \ge \frac{3}{4}$. Moreover $\operatorname{Var} W \le \frac{1}{n}\mathbb{E}[e^{2\lambda\xi-\lambda^2}\mathbf{1}\{\xi \le u\}] = \frac{1}{n}e^{\lambda^2}\mathbb{P}(\xi \le u-2\lambda)$. If $2\lambda \le u$ this is at most $e^{\lambda^2-\ell} \le e^{-\ell/2}$. If $2\lambda > u$, it is at most $e^{\lambda^2-\ell-(2\lambda-u)^2/2} = e^{-(u-\lambda)^2} \le e^{-2\upsilon^2\ell}$. Since $2\upsilon^2 \le \frac{1}{2}$, $\upsilon := \operatorname{Var} W \le 2e^{-2\upsilon^2\ell}$.

On $\{W \ge p/2\}$ we use $\log w \ge (w-1) - (w-1)^2$ for $w \ge \frac{1}{2}$ (the difference of the two sides vanishes at $w=1$, is decreasing on $[\frac{1}{2},1]$ and increasing on $[1,\infty)$) with $w = W/p$. Since $\widehat{W} \ge W$, $\log p \le 0$, $\mathbb{E}(W/p-1)^2 = \upsilon/p^2$ and $\mathbb{E}[(W/p-1)\mathbf{1}\{W \ge p/2\}] = -\mathbb{E}[(W/p-1)\mathbf{1}\{W < p/2\}] \ge 0$, this gives $\mathbb{E}[\log\widehat{W}; W \ge p/2] \ge \log p - \upsilon/p^2$. On $\{W < p/2\}$ we use $\log\widehat{W} \ge \lambda\xi_1 - \lambda^2/2 - \ell$, together with $\mathsf{Q} := \mathbb{P}(W < p/2) \le 4\upsilon/p^2$ (Chebyshev) and Cauchy–Schwarz. This gives $\mathbb{E}[\log\widehat{W}; W < p/2] \ge -(\lambda^2/2+\ell)\mathsf{Q} - \lambda\sqrt{\mathsf{Q}}$. With $p \ge \frac{3}{4}$, $\log p \ge -\frac{4}{3}e^{-\upsilon^2\ell}$, $\upsilon/p^2 \le 3.6e^{-2\upsilon^2\ell}$, $\mathsf{Q} \le 14.3e^{-2\upsilon^2\ell}$ and $\lambda^2 \le 2\ell$, we obtain

$$\frac{\lambda^2}{2} - \psi_n(\lambda) \le \Big(\tfrac{4}{3} + 3.6 + 28.6\,\ell + 5.4\sqrt{\ell}\Big)e^{-\upsilon^2\ell} \le 40(1+\ell)e^{-\upsilon^2\ell}.$$

For the consequence take $\upsilon = \ell^{-1/3}$ and $\ell \ge 1000$, so that $\upsilon \le \frac{1}{2}$ and $e^{-\upsilon^2\ell} \le \frac{1}{4}$. Then $40(1+\ell)e^{-\ell^{1/3}}/\ell \to 0$ and $\upsilon^2\ell/\ell \to 0$, while for $\lambda \le \kappa_0\sqrt{\ell}$ we have $\lambda(\sqrt{2\ell} - \mu_n)/\ell \le \kappa_0(\log\ell + 3)/(\sqrt{2}\,\ell) \to 0$ by Lemma 3.4(b). □

3.5. **Generalised random energy models.** Let $\mathcal{T}$ be a rooted tree of depth m in which every vertex at depth $j-1$ has $n_j \ge 1$ children ($1 \le j \le m$). Write $|v|$ for the depth of v, v_j for its ancestor at depth j, and $\mathcal{T}_j$ for the set of vertices at depth j. Given $\alpha_1,\dots,\alpha_m \ge 0$ and i.i.d. standard Gaussians $(\xi_v)_{v\neq\mathrm{root}}$, the *GREM* with parameters $(n_j,\alpha_j)_{j\le m}$ is the process

$$S_v := \sum_{j=1}^{|v|} \sqrt{\alpha_j}\,\xi_{v_j}, \qquad v \in \mathcal{T}.$$

If the deepest common ancestor of two leaves $v \neq w$ lies at depth $j-1$, then $\mathbb{E}(S_v - S_w)^2 = 2\sum_{i\ge j}\alpha_i$. We write $E_j := \log n_j$ and $\mathcal{E}_j := E_1 + \cdots + E_j$. By Lemma 3.3, $\max_{\mathcal{T}_m} S$ and $\log\sum_{v\in\mathcal{T}_m} w_v e^{\lambda S_v}$ concentrate with Lipschitz constants σ and $\lambda\sigma$, where $\sigma^2 := \sum_j \alpha_j$.

4. Upper bounds: tree majorants and a barrier lemma

4.1. **A barrier lemma for GREMs with concave profile.** Consider a GREM with parameters $(n_j,\alpha_j)_{j\le m}$ as in Section 3.5, where $n_j \ge 2$ and $\alpha_j > 0$, and set

$$\theta_j := \sqrt{2E_j/\alpha_j}, \qquad 1 \le j \le m.$$

We say that the GREM has a *concave profile* if $\theta_1 \le \theta_2 \le \cdots \le \theta_m$, i.e. if the variance per unit of entropy α_j/E_j is non-increasing. Equivalently, the piecewise linear function interpolating the points $(\mathcal{E}_j, \alpha_1+\cdots+\alpha_j)$, $0 \le j \le m$ (with $\mathcal{E}_0 = 0$), is concave. For $\delta > 0$ put

$$c_j := (1+\delta)\theta_j, \qquad m_j := (1+\delta)\sum_{i\le j}\sqrt{2\alpha_i E_i}, \qquad P_\delta := \sum_{j=1}^{m} e^{-(2\delta+\delta^2)\mathcal{E}_j}. \tag{4.1}$$

For a fixed number of levels, the first-order asymptotics behind Corollary 4.2 below is [20, Thm 1.5(iii)], and the corresponding formula for continuous hierarchies is [21, Thm 3.1]. Lemma 4.1 is a non-asymptotic version of the first moment arguments along a barrier that are used for branching random walks in time-inhomogeneous environments with decreasing variances [43, 42, 61, 60, 58]; see also [54, 17].

Lemma 4.1 (Barrier)**.** *For a GREM with concave profile, every $\delta > 0$ and every $u \geq 0$,*

$$\mathbb{P}\Big(\exists\, v \in \mathcal{T},\ 1 \leq |v| \leq m :\ S_v > m_{|v|} + u\Big)\ \leq\ P_\delta\, e^{-c_1 u}.$$

Proof. If the event occurs, choose such a v of minimal depth j. Then $S_v > m_j + u$ and $S_{v_i} \leq m_i + u$ for all $i < j$. Since $\mathcal{T}_j$ has $e^{\mathcal{E}_j}$ vertices and all of them have the same path law, a union bound gives

$$\mathbb{P}(\cdots) \leq \sum_{j=1}^{m} e^{\mathcal{E}_j}\, \mathbb{P}\Big(S_j > m_j + u,\ S_i \leq m_i + u\ \ \forall i < j\Big),$$

where $S_i = \sum_{l \leq i} X_l$ with independent $X_l \sim N(0, \alpha_l)$. Fix j. By Abel summation,

$$\sum_{i \leq j} c_i X_i = c_j S_j - \sum_{i<j} (c_{i+1} - c_i) S_i.$$

Since $c_{i+1} - c_i \geq 0$, on the event in question the right-hand side is at least $c_j(m_j + u) - \sum_{i<j}(c_{i+1} - c_i)(m_i + u) = \sum_{i \leq j} c_i(m_i - m_{i-1}) + c_1 u$, with $m_0 := 0$. Therefore the probability is at most

$$\mathbb{E}\exp\Big(\sum_{i \leq j} c_i X_i\Big)\, e^{-\sum_{i \leq j} c_i (m_i - m_{i-1}) - c_1 u} = \exp\Big(\sum_{i \leq j} \big[\tfrac{1}{2} c_i^2 \alpha_i - c_i(m_i - m_{i-1})\big] - c_1 u\Big).$$

Now $\frac{1}{2} c_i^2 \alpha_i = (1+\delta)^2 E_i$ and $c_i(m_i - m_{i-1}) = 2(1+\delta)^2 E_i$, so the exponent is $-(1+\delta)^2 \mathcal{E}_j - c_1 u$. Multiplying by $e^{\mathcal{E}_j}$ and summing over j gives the claim. □

Corollary 4.2. *For a GREM with concave profile and every $\delta > 0$,*

$$\mathbb{E} \max_{v \in \mathcal{T}_m} S_v\ \leq\ (1+\delta) \sum_{j=1}^{m} \sqrt{2\alpha_j E_j}\ +\ \frac{1 + \log^+ P_\delta}{c_1}.$$

Proof. Let $Y = \max_{\mathcal{T}_m} S$. By Lemma 4.1, $\mathbb{P}(Y > m_m + u) \leq \min(1, P_\delta e^{-c_1 u})$. Hence $\mathbb{E}Y \leq m_m + \int_0^\infty \min(1, P_\delta e^{-c_1 u})\, du \leq m_m + (1 + \log^+ P_\delta)/c_1$. □

The next proposition is a non-asymptotic form of the upper half of the GREM free-energy formula of Derrida and Gardner [35] and Capocaccia, Cassandro and Picco [23]; cf. [20, Thm 1.6].

Proposition 4.3 (GREM free energy, upper bound)**.** *Consider a GREM with concave profile and the uniform weights $p := \prod_{j \leq m} n_j^{-1}$ on $\mathcal{T}_m$. Let $\lambda \geq 0$, $Z := \sum_{v \in \mathcal{T}_m} p\, e^{\lambda S_v}$ and $\sigma^2 := \sum_j \alpha_j$. For every $\delta > 0$,*

$$\mathbb{E}\log Z\ \leq\ \sum_{j=1}^{m} \mathsf{R}\big(\lambda\sqrt{\alpha_j}; E_j\big) + \delta\lambda \sum_{j=1}^{m} \sqrt{2\alpha_j E_j} + \frac{\delta^2 \lambda^2 \sigma^2}{2} + \frac{\lambda\, \log^+(4P_\delta)}{c_1} + \lambda\sigma\sqrt{2\log 2} + 2.$$

Proof. Let $k := \max\{j \in \{0, \dots, m\} : j = 0 \text{ or } c_j \leq \lambda\}$. Since (c_j) is non-decreasing, $c_j \leq \lambda$ exactly for $j \leq k$.

Step 1: a good event. Let $u := \log^+(4P_\delta)/c_1$ and $\Omega_\star := \{S_v \leq m_{|v|} + u \text{ for all } v \text{ with } 1 \leq |v| \leq k\}$, where $\Omega_\star$ is the sure event if $k = 0$. By Lemma 4.1, $\mathbb{P}(\Omega_\star^c) \leq P_\delta e^{-c_1 u} \leq \frac{1}{4}$: if $4P_\delta \geq 1$ then $P_\delta e^{-c_1 u} = \frac{1}{4}$, and otherwise $u = 0$ and $P_\delta < \frac{1}{4}$.

Step 2: a first moment on $\Omega_\star$. For $v \in \mathcal{T}_k$ let $A_v := \{S_{v_i} \leq m_i + u,\ 1 \leq i \leq k\}$ and $W_v := \sum_{w \in \mathcal{T}_m,\, w \geq v} \big(\prod_{i>k} n_i^{-1}\big) e^{\lambda(S_w - S_v)}$. Then $Z = \sum_{v \in \mathcal{T}_k} \big(\prod_{i \leq k} n_i^{-1}\big) e^{\lambda S_v} W_v$, and $Z = Z_\star := \sum_{v \in \mathcal{T}_k} \big(\prod_{i \leq k} n_i^{-1}\big) e^{\lambda S_v} \mathbf{1}_{A_v} W_v$ on $\Omega_\star$. The variable W_v is independent of $(S_{v_i})_{i \leq k}$, and $\mathbb{E}W_v = \exp(\frac{\lambda^2}{2} \sum_{i>k} \alpha_i)$. For $k \geq 1$ set $\gamma_i := c_{i+1} - c_i$ for $i < k$ and $\gamma_k := \lambda - c_k$; all $\gamma_i \geq 0$. Using

$\mathbf{1}_{A_v} \le \exp(\sum_{i\le k}\gamma_i(m_i+u-S_{v_i}))$ and $\lambda S_{v_k} - \sum_{i\le k}\gamma_i S_{v_i} = \sum_{i\le k} c_i X_i$ (Abel summation), we get

$$\mathbb{E}\big[e^{\lambda S_v}\mathbf{1}_{A_v}\big] \le \exp\Big(\sum_{i\le k}\gamma_i(m_i+u) + \sum_{i\le k}\tfrac12 c_i^2\alpha_i\Big).$$

Since $\sum_{i\le k}\gamma_i = \lambda - c_1 \le \lambda$ and $\sum_{i\le k}\gamma_i m_i = \lambda m_k - \sum_{i\le k} c_i(m_i - m_{i-1})$, the computation in the proof of Lemma 4.1 yields

$$\log \mathbb{E} Z_\star \le \lambda m_k + \lambda u - (1+\delta)^2\mathcal{E}_k + \frac{\lambda^2}{2}\sum_{i>k}\alpha_i.$$

This also holds for $k=0$, with $m_0 = \mathcal{E}_0 = 0$.

Step 3: from probability to expectation. By Markov's inequality, $\mathbb{P}(\log Z > \log\mathbb{E}Z_\star + 2) \le \mathbb{P}(\Omega_\star^c) + \mathbb{P}(Z_\star > e^2\mathbb{E}Z_\star) \le \frac14 + e^{-2} < \frac12$. The variable $\log Z$ is a $\lambda\sigma$-Lipschitz function of the underlying Gaussian vector, so by (3.2), $\mathbb{E}\log Z \le \log\mathbb{E}Z_\star + 2 + \lambda\sigma\sqrt{2\log 2}$.

Step 4: comparison with R. For $i \le k$ we have $\lambda \ge c_i \ge \theta_i$, i.e. $\lambda\sqrt{\alpha_i} \ge \sqrt{2E_i}$, and

$$(1+\delta)\lambda\sqrt{2\alpha_i E_i} - (1+\delta)^2 E_i \le \lambda\sqrt{2\alpha_i E_i} - E_i + \delta\lambda\sqrt{2\alpha_i E_i} = \mathsf{R}(\lambda\sqrt{\alpha_i}; E_i) + \delta\lambda\sqrt{2\alpha_i E_i}.$$

For $i > k$ we have $\lambda\sqrt{\alpha_i} < (1+\delta)\sqrt{2E_i}$. If $\lambda\sqrt{\alpha_i} \le \sqrt{2E_i}$ then $\frac12\lambda^2\alpha_i = \mathsf{R}(\lambda\sqrt{\alpha_i}; E_i)$. Otherwise $\frac12\lambda^2\alpha_i - \mathsf{R}(\lambda\sqrt{\alpha_i}; E_i) = \frac12(\lambda\sqrt{\alpha_i} - \sqrt{2E_i})^2 \le \frac12\big(\frac{\delta}{1+\delta}\lambda\sqrt{\alpha_i}\big)^2 \le \frac12\delta^2\lambda^2\alpha_i$. Since $\lambda m_k - (1+\delta)^2\mathcal{E}_k = \sum_{i\le k}[(1+\delta)\lambda\sqrt{2\alpha_i E_i} - (1+\delta)^2 E_i]$, combining Steps 2–4 proves the proposition. □

4.2. **Tree majorants.** For $j \ge 0$ let $\mathcal{Q}_j$ be the partition of $\mathsf{T} = [0,1]^d$ into the b^{dj} cubes $\prod_{i=1}^d I_{j,k_i}$, where $I_{j,k} = [kb^{-j}, (k+1)b^{-j})$ for $0 \le k < b^j - 1$ and $I_{j,b^j-1} = [1-b^{-j}, 1]$. Each cube of $\mathcal{Q}_j$ is the union of b^d cubes of $\mathcal{Q}_{j+1}$. Ordering cubes by inclusion turns $\bigcup_{j\le K}\mathcal{Q}_j$ into a rooted tree of depth K with constant branching number b^d. For $Q \in \mathcal{Q}_K$ let t_Q be its lower-left corner, and for a process X let $\omega_X(r) := \sup\{|X_t - X_s| : s,t \in \mathsf{T},\ |t-s| \le r\}$.

Proposition 4.4 (Tree majorant)**.** *Let X be a centred Gaussian process on T with continuous paths, and let $\varphi : (0,1] \to [0,\infty)$ be non-decreasing with $\mathsf{d}_X(s,t)^2 \le \varphi(|t-s|)$ for $s \ne t$. For integers $b \ge 2$, $K \ge 1$ let $\alpha_j := \frac12(\varphi(b^{-j+1}) - \varphi(b^{-j}))$, $1 \le j \le K$, and let S be the GREM with parameters $(b^d, \alpha_j)_{j\le K}$ on the tree of b-adic cubes. Then*

$$\mathbb{E}\sup_{\mathsf{T}} X \le \mathbb{E}\max_{Q\in\mathcal{Q}_K} S_Q + \sqrt{\varphi(b^{-K})\,dK\log b} + \mathbb{E}\,\omega_X(b^{-K}), \tag{4.2}$$

$$\mathbb{E}\log\int_{\mathsf{T}} e^{\lambda X_t}\,dt \le \mathbb{E}\log\sum_{Q\in\mathcal{Q}_K} b^{-dK}e^{\lambda S_Q} + \lambda\sqrt{\varphi(b^{-K})\,dK\log b} + \lambda\,\mathbb{E}\,\omega_X(b^{-K}), \quad \lambda \ge 0. \tag{4.3}$$

Moreover $\mathbb{E}\,\omega_X(r) \to 0$ as $r \downarrow 0$.

Proof. Let $(\xi'_Q)_{Q\in\mathcal{Q}_K}$ be i.i.d. standard Gaussians, independent of S, and $Y_Q := S_Q + \sqrt{\nu}\,\xi'_Q$ with $\nu := \frac12\varphi(b^{-K})$. Let $Q \ne Q'$ in $\mathcal{Q}_K$ have deepest common ancestor at depth $j-1 \in \{0,\dots,K-1\}$. Both corners $t_Q, t_{Q'}$ lie in one cube of $\mathcal{Q}_{j-1}$, so $|t_Q - t_{Q'}| \le b^{-j+1}$. Hence

$$\mathbb{E}(Y_Q - Y_{Q'})^2 = 2\sum_{i=j}^K \alpha_i + 2\nu = \varphi(b^{-j+1}) \ \ge\ \varphi(|t_Q - t_{Q'}|) \ \ge\ \mathsf{d}_X(t_Q, t_{Q'})^2.$$

By Corollary 3.2, $\mathbb{E}\max_Q X_{t_Q} \le \mathbb{E}\max_Q Y_Q \le \mathbb{E}\max_Q S_Q + \sqrt{\nu}\,\mathbb{E}\max_Q \xi'_Q$. By Lemma 3.4(a), $\sqrt{\nu}\,\mathbb{E}\max_Q\xi'_Q \le \sqrt{2\nu\,dK\log b}$. Since every $t \in \mathsf{T}$ lies in some $Q \in \mathcal{Q}_K$ with $|t - t_Q| \le b^{-K}$, we

have $\sup_{\mathsf{T}} X \le \max_Q X_{t_Q} + \omega_X(b^{-K})$, which gives (4.2). Similarly

$$\int_{\mathsf{T}} e^{\lambda X_t} dt = \sum_Q \int_Q e^{\lambda X_t} dt \le e^{\lambda \omega_X(b^{-K})} \sum_Q b^{-dK} e^{\lambda X_{t_Q}},$$

and Lemma 3.1 with weights b^{-dK} gives

$$\mathbb{E} \log \sum_Q b^{-dK} e^{\lambda X_{t_Q}} \le \mathbb{E} \log \sum_Q b^{-dK} e^{\lambda Y_Q} \le \mathbb{E} \log \sum_Q b^{-dK} e^{\lambda S_Q} + \lambda \sqrt{\nu}\, \mathbb{E} \max_Q \xi'_Q,$$

which gives (4.3). Finally, since X has continuous paths, $\sup_{\mathsf{T}} |X| = \sup_D |X|$ for a countable dense set $D \subset \mathsf{T}$, and this is almost surely finite; Borell–TIS on D [1, Thm. 2.1.1] (see also [44]) gives $\mathbb{E} \sup_{\mathsf{T}} |X| < \infty$. Since $\omega_X(r) \le 2 \sup |X|$ and $\omega_X(r) \to 0$ almost surely by uniform continuity, dominated convergence gives $\mathbb{E}\, \omega_X(r) \to 0$. □

4.3. **Proof of the upper bounds for fractional Brownian motion.**

Proof of Theorem 2.1(i). Apply Proposition 4.4 with $d = 1$ and $\varphi(r) = r^{2H}$. Then $\alpha_j = \frac{1}{2}(1 - b^{-2H})\, b^{-2H(j-1)}$, which is strictly decreasing in j, so the GREM has a concave profile. By Corollary 4.2,

$$\mathbb{E} \max_{\mathcal{Q}_K} S \le (1+\delta)\sqrt{2 \log b} \sum_{j=1}^{\infty} \sqrt{\alpha_j} + \frac{1 + \log^+ P_\delta}{c_1},$$

with $P_\delta \le \sum_{j \ge 1} b^{-j(2\delta + \delta^2)} = P_{b,\delta}$ and $c_1 = (1+\delta)\sqrt{2 \log b / \alpha_1} = c_{b,\delta,H}$. Moreover

$$\sqrt{2 \log b} \sum_{j \ge 1} \sqrt{\alpha_j} = \frac{\sqrt{(1 - b^{-2H}) \log b}}{1 - b^{-H}} = \sqrt{\log b \, \frac{1 + b^{-H}}{1 - b^{-H}}} = \sqrt{\log b \, \coth\bigl(\tfrac{1}{2} H \log b\bigr)}.$$

The remaining two terms in (4.2) tend to zero as $K \to \infty$, since $K b^{-2HK} \to 0$ and by Proposition 4.4. This proves (2.1). □

Proof of the upper bound in Theorem 2.1(iii). Let $H \in (0, \frac{1}{2}]$ and take $b = 2$, $\delta = H$ in (2.1). From $\coth x \le x^{-1} + x/3$ for $x > 0$ (which follows from $\coth x - x^{-1} = \sum_{k \ge 1} 2x/(x^2 + k^2 \pi^2) \le (2x/\pi^2) \sum_k k^{-2}$) we get $\log 2 \, \coth(\frac{1}{2} H \log 2) \le \frac{2}{H} + \frac{H \log^2 2}{6}$, hence

$$(1+H)\sqrt{\log 2 \, \coth(\tfrac{1}{2} H \log 2)} \le (1+H)\Bigl(\sqrt{2/H} + \frac{H^{3/2} \log^2 2}{12\sqrt{2}}\Bigr) \le \sqrt{2/H} + \sqrt{2H} + 0.05\, H^{3/2}.$$

Next, $2^{2H+H^2} - 1 \ge (2H + H^2) \log 2 \ge 2H \log 2$, so $\log^+ P_{2,H} \le \log^+ \frac{1}{2H \log 2} \le \log(1/H)$, using $2 \log 2 > 1$ and $\log(1/H) \ge 0$. Also $1 - 2^{-2H} \le 2H \log 2$ gives $c_{2,H,H} \ge \sqrt{2/H}$. Hence the second term in (2.1) is at most $\sqrt{H/2}\,(1 + \log(1/H))$, and

$$M_H - \sqrt{2/H} \le \sqrt{H}\Bigl(\sqrt{2} + 0.025 + \tfrac{1}{\sqrt{2}} + \tfrac{1}{\sqrt{2}} \log(1/H)\Bigr) \le \sqrt{H}\bigl(2 + \log(1/H)\bigr),$$

where we used $0.05 H^{3/2} \le 0.025 \sqrt{H}$ and, in the last step, $(1 - 2^{-1/2}) \log(1/H) \ge (1 - 2^{-1/2}) \log 2 > \sqrt{2} + 2^{-1/2} + 0.025 - 2$. □

4.4. **Proof of the universal upper bounds.**

Proposition 4.5. *Let (X^ε) be centred Gaussian processes on T with continuous paths that satisfy (2.4) and (2.6), for some $r_\varepsilon \in (0,1]$ with $\varepsilon \log(1/r_\varepsilon) \to 0$. Then*

$$\limsup_{\varepsilon\downarrow 0} \sqrt{\varepsilon}\,\mathbb{E}\sup_{\mathsf{T}} X^\varepsilon \le \sqrt{d}\, I_\Phi \quad \text{and} \quad \limsup_{\varepsilon\downarrow 0} \varepsilon\,\mathbb{E}\log \int_{\mathsf{T}} e^{\beta X_t^\varepsilon/\sqrt{\varepsilon}} dt \le F_{\Phi,d}(\beta) \quad (\beta \ge 0).$$

Proof. Step 1: the case $r_\varepsilon = 1$. Fix $b = 2$ and $\delta \in (0,1)$, and abbreviate $\eta = \eta(\varepsilon)$, $h := \varepsilon \log b$, $y_j := jh$, $g = -\Phi'$. Apply Proposition 4.4 with $\varphi(r) := (1+\eta)\Phi(\varepsilon\log(1/r))$, which is non-decreasing in r, so that $\varphi(b^{-j}) = (1+\eta)\Phi(y_j)$ and

$$\alpha_j = \frac{1+\eta}{2}\int_{y_{j-1}}^{y_j} g \;\in\; \Big[\frac{(1+\eta)h}{2} g(y_j),\ \frac{(1+\eta)h}{2} g(y_{j-1})\Big]. \tag{4.4}$$

Since g is positive and non-increasing, (α_j) is positive and non-increasing, so the GREM has a concave profile ($E_j = d\log b$ for all j). The error term $\varphi(b^{-K}) dK \log b = (1+\eta)\, d\,\varepsilon^{-1} y_K \Phi(y_K)$ tends to 0 as $K \to \infty$ by (2.3), and so does $\mathbb{E}\,\omega_{X^\varepsilon}(b^{-K})$. In both bounds below we may therefore let $K \to \infty$.

Maximum. Since $\sqrt{g}$ is non-increasing, $\sum_{j\ge 1}\sqrt{g(y_{j-1})} \le \sqrt{g(0)} + h^{-1} I_\Phi$, so by (4.4)

$$\sqrt{2d\log b}\sum_{j\ge 1}\sqrt{\alpha_j} \le \sqrt{(1+\eta)\,d\,h\log b}\,\Big(\sqrt{g(0)} + \frac{I_\Phi}{h}\Big) = \sqrt{\frac{(1+\eta)d}{\varepsilon}}\big(I_\Phi + h\sqrt{g(0)}\big). \tag{4.5}$$

Moreover $\alpha_1 \le \frac{1}{2}(1+\eta)hg(0)$ gives $c_1 \ge (1+\delta)\sqrt{4d/((1+\eta)\varepsilon g(0))}$, and $P_\delta \le (b^{2d\delta}-1)^{-1} \le (2d\delta\log b)^{-1}$. By Proposition 4.4 and Corollary 4.2,

$$\sqrt{\varepsilon}\,\mathbb{E}\sup_{\mathsf{T}} X^\varepsilon \le (1+\delta)\sqrt{(1+\eta)d}\,\big(I_\Phi + h\sqrt{g(0)}\big) + \varepsilon\sqrt{\frac{(1+\eta)g(0)}{4d}}\Big(1 + \log^+\frac{1}{2d\delta\log 2}\Big).$$

Letting $\varepsilon \downarrow 0$ and then $\delta \downarrow 0$ proves the first claim when $r_\varepsilon = 1$.

Free energy. Let $\lambda = \beta/\sqrt{\varepsilon}$. By (4.4) and the monotonicity of R in α, $\mathsf{R}(\lambda\sqrt{\alpha_j}; d\log b) \le \mathsf{R}(\lambda\sqrt{\alpha_j^+}; d\log b)$ with $\alpha_j^+ := \frac{1}{2}(1+\eta)hg(y_{j-1})$. Writing $\mathsf{z}_j := \beta\sqrt{(1+\eta)g(y_{j-1})}$ we have $\lambda\sqrt{\alpha_j^+} = \mathsf{z}_j\sqrt{\log b/2}$, and a direct check from (3.4) and (2.7) gives the key identity

$$\mathsf{R}\big(z\sqrt{\log b/2};\, d\log b\big) = \log b\; G_d(z), \qquad z \ge 0. \tag{4.6}$$

Since $y \mapsto G_d(\beta\sqrt{(1+\eta)g(y)})$ is non-increasing,

$$\varepsilon\sum_{j\ge 1}\mathsf{R}(\lambda\sqrt{\alpha_j}; d\log b) \le \sum_{j\ge 1} h\, G_d(\mathsf{z}_j) \le h\, G_d\big(\beta\sqrt{(1+\eta)g(0)}\big) + F_{\Phi,d}\big(\beta\sqrt{1+\eta}\big).$$

The remaining terms of Proposition 4.3, multiplied by ε, are bounded as follows. Using (4.5),

$$\varepsilon\delta\lambda\sum_j\sqrt{2\alpha_j d\log b} \le \delta\beta\sqrt{(1+\eta)d}\,\big(I_\Phi + h\sqrt{g(0)}\big).$$

Since $\sigma^2 = \sum_j \alpha_j \le \frac{1}{2}(1+\eta)\Phi(0)$, we get $\varepsilon\delta^2\lambda^2\sigma^2/2 \le \frac{1}{4}\delta^2\beta^2(1+\eta)\Phi(0)$. The terms $\varepsilon\lambda\log^+(4P_\delta)/c_1$, $\varepsilon\lambda\sigma\sqrt{2\log 2}$ and 2ε are $O_\delta(\sqrt{\varepsilon})$. Together with (4.3) and the continuity of the convex function $F_{\Phi,d}$, this gives

$$\limsup_{\varepsilon\downarrow 0}\varepsilon\,\mathbb{E}\log\int_{\mathsf{T}} e^{\beta X^\varepsilon/\sqrt{\varepsilon}} \le F_{\Phi,d}(\beta) + \delta\beta\sqrt{d}\, I_\Phi + \tfrac{1}{4}\delta^2\beta^2\Phi(0),$$

and $\delta \downarrow 0$ proves the second claim when $r_\varepsilon = 1$.

Every bound in Step 1 holds for each centred Gaussian process X on T with continuous paths that satisfies (2.4) with $r_\varepsilon = 1$, and its right-hand side depends on X only through Φ, η, d, ε and δ. Taking the infimum over $\delta \in (0,1)$, we obtain numbers U_ε and $V_\varepsilon(\beta)$, depending only on Φ, η, d, ε and β, such that $\limsup_{\varepsilon\downarrow 0} \sqrt{\varepsilon}\, U_\varepsilon \le \sqrt{d}\, I_\Phi$, $\limsup_{\varepsilon\downarrow 0} \varepsilon\, V_\varepsilon(\beta) \le F_{\Phi,d}(\beta)$, and

$$\mathbb{E} \sup_{\mathsf{T}} X \le U_\varepsilon \qquad \text{and} \qquad \mathbb{E} \log \int_{\mathsf{T}} e^{\beta X_t/\sqrt{\varepsilon}} dt \le V_\varepsilon(\beta) \tag{4.7}$$

for every such X.

Step 2: the general case. Let $J := \lceil \log_2(1/r_\varepsilon) \rceil$, so that $2^{-J} \le r_\varepsilon$ and $\varepsilon J \to 0$. Let $\mathcal{Q}_J$ be the partition of T into the $N := 2^{dJ}$ dyadic cubes of side 2^{-J} defined above (with $b = 2$). For $Q \in \mathcal{Q}_J$ let t_Q be its lower-left corner and $X^Q_u := X^\varepsilon_{t_Q + 2^{-J}u} - X^\varepsilon_{t_Q}$ for $u \in \mathsf{T}$. If $u \ne v \in \mathsf{T}$, then $|2^{-J}(u-v)| \le r_\varepsilon$ and $\varepsilon \log(2^J/|u-v|) \ge \varepsilon \log(1/|u-v|)$, so by (2.4) and the monotonicity of Φ, the process X^Q satisfies (2.4) with $r_\varepsilon = 1$. It has continuous paths, and $\operatorname{Var} X^Q_u \le (1+\eta)\Phi(0) \le 2\Phi(0) =: \sigma_0^2$. Fix $t_* \in \mathsf{T}$. By (2.6) the N variables $X^\varepsilon_{t_Q} - X^\varepsilon_{t_*}$ have variances at most C_0, so Lemma 3.4(a) gives

$$\mathbb{E} \max_{Q \in \mathcal{Q}_J} X^\varepsilon_{t_Q} = \mathbb{E} \max_{Q \in \mathcal{Q}_J} \bigl(X^\varepsilon_{t_Q} - X^\varepsilon_{t_*} \bigr) \le \sqrt{2C_0 \log N} = \sqrt{2C_0\, dJ \log 2}.$$

Maximum. Since every $t \in \mathsf{T}$ lies in the closure of some $Q \in \mathcal{Q}_J$, we have $\sup_{\mathsf{T}} X^\varepsilon \le \max_Q X^\varepsilon_{t_Q} + \max_Q Y_Q$ with $Y_Q := \sup_{\mathsf{T}} X^Q$. By (4.7), $\mathbb{E} Y_Q \le U_\varepsilon$, and by Borell–TIS on a countable dense set, the variables Y_Q satisfy the hypothesis of (3.3) with $s = \sigma_0$. Hence

$$\sqrt{\varepsilon}\, \mathbb{E} \sup_{\mathsf{T}} X^\varepsilon \le \sqrt{2C_0\, \varepsilon\, dJ \log 2} + \sqrt{\varepsilon}\, U_\varepsilon + \sigma_0 \sqrt{\varepsilon} \Bigl(1 + \sqrt{2 \log(2N)} \Bigr).$$

The first and the last term tend to zero because $\varepsilon J \to 0$, which proves the first claim.

Free energy. Let $\lambda = \beta/\sqrt{\varepsilon}$. The change of variables $t = t_Q + 2^{-J}u$ gives $\int_Q e^{\lambda X^\varepsilon_t} dt = 2^{-dJ} e^{\lambda X^\varepsilon_{t_Q}} e^{f_Q}$ with $f_Q := \log \int_{\mathsf{T}} e^{\lambda X^Q_u} du$. Summing over Q, we get $\log \int_{\mathsf{T}} e^{\lambda X^\varepsilon} \le \lambda \max_Q X^\varepsilon_{t_Q} + \max_Q f_Q$. By (4.7), $\mathbb{E} f_Q \le V_\varepsilon(\beta)$, and by Lemma 3.3(c) the variables f_Q satisfy the hypothesis of (3.3) with $s = \lambda\sigma_0$. Hence

$$\varepsilon\, \mathbb{E} \log \int_{\mathsf{T}} e^{\beta X^\varepsilon_t/\sqrt{\varepsilon}} dt \le \beta \sqrt{2C_0\, \varepsilon\, dJ \log 2} + \varepsilon\, V_\varepsilon(\beta) + \beta\sigma_0 \sqrt{\varepsilon} \Bigl(1 + \sqrt{2 \log(2N)} \Bigr),$$

and the second claim follows in the same way. □

5. Lower bounds: Cantor minorants

For lower bounds the comparison has to go the other way. We need a tree whose increments are *dominated* by those of X. Points in different branches of a b-adic tree can be arbitrarily close, so we use a Cantor-type subtree whose branches are separated by at least the size of the cells.

5.1. **Cantor cells.** Let $Q_0 = a + [0,q]^d$ be a closed cube and $b \ge 2$ an integer. Set $\mathcal{C}_0 := \{Q_0\}$. Given a closed cube $C \in \mathcal{C}_{j-1}$ of side r_{j-1}, divide it into b^d closed congruent subcubes of side r_{j-1}/b, and inside each of them take the concentric closed cube of side

$$r_j := \frac{r_{j-1}}{2b} = q\,(2b)^{-j}.$$

These b^d cubes are the *children* of C, and $\mathcal{C}_j$ is the set of all children of cells in $\mathcal{C}_{j-1}$. This defines a rooted tree of depth K with constant branching number b^d for every K. Each cell of $\mathcal{C}_j$ has volume r_j^d, and the cells of $\mathcal{C}_j$ have total volume $q^d 2^{-dj}$.

Lemma 5.1 (Separation)**.** *If C', C'' are distinct children of the same cell of $\mathcal{C}_{j-1}$, then $|t-s| \geq r_j$ for all $t \in C'$, $s \in C''$.*

Proof. Let $r = r_{j-1}$. Choose a coordinate in which the two subcubes of side r/b containing C' and C'' differ, and let a' be that coordinate of the lower-left corner of the parent cell. In this coordinate the projections of C' and C'' are $a' + [(k+\frac{1}{4})\frac{r}{b}, (k+\frac{3}{4})\frac{r}{b}]$ and $a' + [(k'+\frac{1}{4})\frac{r}{b}, (k'+\frac{3}{4})\frac{r}{b}]$ with $k < k'$. Since $|\cdot|$ is the sup-norm, $|t-s|$ is at least the distance between these projections, which is at least $(k'-k-\frac{1}{2})\frac{r}{b} \geq \frac{r}{2b} = r_j$. □

Lemma 5.2 (Cantor minorant)**.** *Let X be a centred Gaussian process on $\mathsf{T} \supseteq Q_0$, and let $\underline{\varphi} : (0,1] \to [0,\infty)$ be non-decreasing with $\mathsf{d}_X(s,t)^2 \geq \underline{\varphi}(|t-s|)$ for $s \neq t \in Q_0$. Let $K \geq 1$, pick $t_C \in C$ for every $C \in \mathcal{C}_K$, and let S' be the GREM with parameters $(b^d, \alpha'_j)_{j \leq K}$ on the Cantor tree, where*

$$\alpha'_j := \tfrac{1}{2}\big(\underline{\varphi}(r_j) - \underline{\varphi}(r_{j+1})\big) \quad (1 \leq j < K), \qquad \alpha'_K := \tfrac{1}{2}\underline{\varphi}(r_K).$$

Then $\mathbb{E}(S'_C - S'_{C'})^2 \leq \mathsf{d}_X(t_C, t_{C'})^2$ for all $C, C' \in \mathcal{C}_K$. Consequently $\mathbb{E}\max_{C \in \mathcal{C}_K} X_{t_C} \geq \mathbb{E}\max_{C \in \mathcal{C}_K} S'_C$ and, for all weights $w_C > 0$ and $\lambda \geq 0$, $\mathbb{E}\log\sum_C w_C e^{\lambda X_{t_C}} \geq \mathbb{E}\log\sum_C w_C e^{\lambda S'_C}$.

Proof. If the deepest common ancestor of $C \neq C'$ lies at depth $j-1$, then C, C' are contained in distinct children of that ancestor. By Lemma 5.1, $|t_C - t_{C'}| \geq r_j$, hence $\mathsf{d}_X(t_C, t_{C'})^2 \geq \underline{\varphi}(r_j) = 2\sum_{i=j}^K \alpha'_i = \mathbb{E}(S'_C - S'_{C'})^2$. The two consequences follow from Corollary 3.2 and Lemma 3.1. □

5.2. **Greedy and level-wise Jensen bounds for GREMs.**

Proposition 5.3 (Greedy bound)**.** *For a GREM with parameters $(n_j, \alpha_j)_{j \leq m}$ and arbitrary $\alpha_j \geq 0$,*

$$\mathbb{E}\max_{v \in \mathcal{T}_m} S_v \;\geq\; \sum_{j=1}^m \mu_{n_j}\sqrt{\alpha_j}.$$

Proof. Define $v^{(0)}$ to be the root and, recursively, $v^{(j)}$ to be the child of $v^{(j-1)}$ with the largest ξ; ties have probability zero. Given $v^{(0)}, \dots, v^{(j-1)}$, the variables ξ_c over the children c of $v^{(j-1)}$ are i.i.d. standard Gaussians, so $\mathbb{E}\xi_{v^{(j)}} = \mu_{n_j}$. Hence $\mathbb{E}\max S \geq \mathbb{E}S_{v^{(m)}} = \sum_j \sqrt{\alpha_j}\,\mu_{n_j}$. □

The same greedy argument underlies the proof of the lower bound in [20, Thm 1.5(iii)]. For a GREM with concave profile and large branching numbers it is sharp to first order, by Corollary 4.2.

Proposition 5.4 (Level-wise Jensen bound)**.** *Consider a GREM with parameters $(n_j, \alpha_j)_{j \leq m}$, numbers $p_1, \dots, p_m > 0$, and the leaf weights $w := \prod_{j \leq m} p_j$. For $\lambda \geq 0$,*

$$\mathbb{E}\log\sum_{v \in \mathcal{T}_m} w\, e^{\lambda S_v} \;\geq\; \sum_{j=1}^m \Big[\log(n_j p_j) + \psi_{n_j}\big(\lambda\sqrt{\alpha_j}\big)\Big],$$

with ψ_n as in Lemma 3.5 and $\psi_1 := 0$. Moreover $\psi_n \geq 0$.

Proposition 5.4 is a non-asymptotic form of the lower half of the GREM free-energy formula of Derrida and Gardner [35] and Capocaccia, Cassandro and Picco [23]; cf. [20, Thm 1.6] and, for continuous hierarchies, [21, Thm 3.3].

Proof. For v at depth j let $Z_v := \sum_{u\in\mathcal{T}_m,\,u\ge v}\big(\prod_{i>j} p_i\big)e^{\lambda(S_u-S_v)}$, so that the quantity on the left is $\mathbb{E}\log Z_{\mathrm{root}}$. If v has depth $j-1$ and children $c_1,\dots,c_n$ with $n=n_j$, then

$$Z_v = n\,p_j\cdot\frac{1}{n}\sum_{i=1}^{n} e^{\lambda\sqrt{\alpha_j}\xi_{c_i}}Z_{c_i} \quad\text{and}\quad \log Z_v \ge \log(np_j) + \log\Big(\frac{1}{n}\sum_i e^{\lambda\sqrt{\alpha_j}\xi_{c_i}}\Big) + \sum_i q_i\log Z_{c_i},$$

where $q_i = e^{\lambda\sqrt{\alpha_j}\xi_{c_i}}/\sum_k e^{\lambda\sqrt{\alpha_j}\xi_{c_k}}$, by concavity of the logarithm. The vector (q_i) is independent of (Z_{c_i}), the Z_{c_i} are identically distributed, and $\mathbb{E}|\log Z_c| < \infty$ since $|\log Z_c| \le \lambda\max_{u\ge c}|S_u - S_c| + \sum_i|\log p_i| + \log|\mathcal{T}_m|$. Hence $\mathbb{E}\sum_i q_i\log Z_{c_i} = \mathbb{E}\log Z_{c_1}$, and $\mathbb{E}\log Z_v \ge \log(np_j) + \psi_n(\lambda\sqrt{\alpha_j}) + \mathbb{E}\log Z_{c_1}$. Iterating from the root down to the leaves, where $Z=1$, gives the claim. Finally, $\psi_n \ge 0$ by Jensen's inequality, $\log\frac{1}{n}\sum e^{x_i} \ge \frac{1}{n}\sum x_i$. □

5.3. **Lower bounds for fractional Brownian motion.**

Proof of Theorem 2.1(ii). Apply Lemma 5.2 with $d=1$, $Q_0=[0,1]$ and $\underline{\varphi}(r) = r^{2H}$, so $r_j = (2b)^{-j}$. By Proposition 5.3 and $\alpha'_K \ge \frac{1}{2}(r_K^{2H} - r_{K+1}^{2H})$,

$$M_H \ge \mathbb{E}\max_{C\in\mathcal{C}_K} B^H_{t_C} \ge \mu_b\sum_{j=1}^{K}\sqrt{\tfrac{1}{2}\big(r_j^{2H} - r_{j+1}^{2H}\big)}.$$

Letting $K\to\infty$ and writing $x = H\log(2b)$, the sum converges to

$$\sqrt{\tfrac{1}{2}(1-e^{-2x})}\;\frac{e^{-x}}{1-e^{-x}} = e^{-x}\sqrt{\frac{1+e^{-x}}{2(1-e^{-x})}} = (2b)^{-H}\sqrt{\tfrac{1}{2}\coth(x/2)},$$

which is (2.2). The second inequality in (2.2) follows from $\coth(x/2)\ge 2/x$. □

Proof of the lower bound in Theorem 2.1(iii). Let $H \le 3\cdot 10^{-6}$, $L := \frac{1}{2}\sqrt{\log(1/H)/H}$ and $b := \lceil e^L\rceil$, so that $\ell := \log b \in [L, L+1]$. Since $H \le 3\cdot 10^{-6}$ we have $L \ge 1000$ and $\log(1/H)\ge 12$. By (2.2) and Lemma 3.4(b),

$$\sqrt{H}\,M_H \;\ge\; e^{-H(\ell+\log 2)}\sqrt{\frac{2\ell}{\ell+\log 2}}\Big(1-\frac{\log\ell+3}{2\ell}\Big) \;\ge\; \sqrt{2}\Big(1 - H(L+2) - \frac{\log L+4}{2L}\Big).$$

Here we used $e^{-x}\ge 1-x$ together with $\ell\le L+1$ and $\log 2\le 1$; $(1+\log 2/\ell)^{-1/2}\ge 1-\log 2/(2\ell)$; the inequality $(1-a)(1-c)(1-e)\ge 1-a-c-e$ for $a,c,e\in[0,1]$, which gives $(1-\frac{\log 2}{2\ell})(1-\frac{\log\ell+3}{2\ell})\ge 1-\frac{\log\ell+4}{2\ell}$; and the fact that $(\log x+4)/x$ is decreasing for $x\ge 1$, together with $\ell\ge L$. Now $HL = \frac{1}{2}\sqrt{H\log(1/H)}$, and $\log L \le \frac{1}{2}\log(1/H) + \frac{1}{2}\log\log(1/H) \le \log(1/H)$, so

$$\frac{\log L+4}{2L} \le \sqrt{H\log(1/H)} + \frac{4\sqrt{H}}{\sqrt{\log(1/H)}}.$$

Since $\log(1/H)\ge 12$, we get $2H + 4\sqrt{H}/\sqrt{\log(1/H)} \le \frac{1}{2}\sqrt{H\log(1/H)}$, and altogether $\sqrt{H}M_H \ge \sqrt{2}(1-2\sqrt{H\log(1/H)})$. This is the claim. Together with the upper bound, it gives $\sqrt{H}M_H\to\sqrt{2}$. □

5.4. **Universal lower bounds.**

Proposition 5.5. *Let (X^ε) satisfy Assumption 2.5. Then*

$$\liminf_{\varepsilon\downarrow 0}\sqrt{\varepsilon}\,\mathbb{E}\sup_{\mathsf{T}} X^\varepsilon \ge \sqrt{d}\,I_\Phi \quad\textit{and}\quad \liminf_{\varepsilon\downarrow 0}\varepsilon\,\mathbb{E}\log\int_{\mathsf{T}} e^{\beta X^\varepsilon_t/\sqrt{\varepsilon}}dt \ge F_{\Phi,d}(\beta)\quad(\beta\ge 0).$$

Proof. Fix an integer $b \geq 2$. Apply the Cantor construction in the cube Q_ε of Assumption 2.5, of side $q := q_\varepsilon$, with $\underline{\varphi}(r) := (1-\eta)\Phi(\varepsilon \log(1/r))$, where $\eta = \eta(\varepsilon)$. Put $h' := \varepsilon \log(2b)$ and $z_j := \varepsilon \log(1/r_j) = \varepsilon \log(1/q) + jh'$. By (2.5) and Lemma 5.2, the GREM S' with

$$\alpha'_j = \frac{1-\eta}{2} \int_{z_j}^{z_{j+1}} g \;\in \Big[\frac{(1-\eta)h'}{2} g(z_{j+1}),\; \frac{(1-\eta)h'}{2} g(z_j) \Big] \quad (j < K), \qquad \alpha'_K = \frac{1-\eta}{2} \Phi(z_K), \tag{5.1}$$

is dominated by X^ε on the points t_C.

Maximum. By Proposition 5.3, (5.1) and the monotonicity of $\sqrt{g}$,

$$\mathbb{E} \sup_{\mathsf{T}} X^\varepsilon \geq \mu_{b^d} \sum_{j=1}^{K-1} \sqrt{\alpha'_j} \geq \mu_{b^d} \sqrt{\frac{(1-\eta)h'}{2}} \sum_{j=1}^{K-1} \sqrt{g(z_{j+1})} \geq \mu_{b^d} \sqrt{\frac{1-\eta}{2h'}} \int_{z_2}^{z_K} \sqrt{g}.$$

Let $K \to \infty$, use $\int_0^{z_2} \sqrt{g} \leq z_2 \sqrt{g(0)} \to 0$ as $\varepsilon \to 0$, and multiply by $\sqrt{\varepsilon}$:

$$\liminf_{\varepsilon \downarrow 0} \sqrt{\varepsilon}\, \mathbb{E} \sup_{\mathsf{T}} X^\varepsilon \geq \frac{\mu_{b^d}}{\sqrt{2 \log(2b)}}\, I_\Phi.$$

Since $\mu_{b^d}/\sqrt{2\log(2b)} \to \sqrt{d}$ as $b \to \infty$ by Lemma 3.4(c), the first claim follows.

Free energy. Fix in addition $Y > 0$, and let $K := \lceil Y/h' \rceil$, so that $z_K \geq Y$ and $\varepsilon K \leq (Y + h')/\log(2b)$. Let $\lambda = \beta/\sqrt{\varepsilon}$, let t_C be the centre of $C \in \mathcal{C}_K$, and $D_C := |C|^{-1} \int_C (X^\varepsilon_t - X^\varepsilon_{t_C}) dt$. By Jensen's inequality on each cell,

$$\int_{\mathsf{T}} e^{\lambda X^\varepsilon_t} dt \;\geq\; \sum_{C \in \mathcal{C}_K} \int_C e^{\lambda X^\varepsilon_t} dt \;\geq\; \sum_{C \in \mathcal{C}_K} |C|\, e^{\lambda X^\varepsilon_{t_C} + \lambda D_C},$$

so

$$\log \int_{\mathsf{T}} e^{\lambda X^\varepsilon} \;\geq\; \log \sum_C |C| e^{\lambda X^\varepsilon_{t_C}} - \lambda \max_C |D_C|.$$

By Fubini and Jensen, $\operatorname{Var} D_C \leq |C|^{-1} \int_C \mathbb{E}(X^\varepsilon_t - X^\varepsilon_{t_C})^2 dt$. For $t \in C \setminus \{t_C\}$ we have $|t - t_C| \leq r_K/2 \leq q_\varepsilon \leq r_\varepsilon$, hence $\varepsilon \log(1/|t-t_C|) \geq \varepsilon \log(2/r_K) \geq z_K$. By (2.4) and the monotonicity of Φ each integrand is at most $(1+\eta)\Phi(z_K)$, so $\operatorname{Var} D_C \leq (1+\eta)\Phi(z_K) \leq (1+\eta)\Phi(Y)$. Lemma 3.4(a), applied to the $2b^{dK}$ variables $\pm D_C$, gives

$$\varepsilon\lambda\, \mathbb{E} \max_C |D_C| \leq \beta \sqrt{2(1+\eta)\Phi(Y)\varepsilon(dK \log b + \log 2)} \leq \beta \sqrt{2(1+\eta)\Phi(Y)(d(Y+h') + \varepsilon \log 2)}.$$

Next, $|C| = q^d \prod_{j \leq K} (2b)^{-d}$, so Lemma 5.2 and Proposition 5.4 (with $p_j = (2b)^{-d}$, $n_j p_j = 2^{-d}$) give

$$\mathbb{E} \log \sum_C |C| e^{\lambda X^\varepsilon_{t_C}} \geq d \log q - dK \log 2 + \sum_{j=1}^{K-1} \psi_{b^d}\big(\lambda \sqrt{\alpha'_j}\big),$$

where we used $\psi_{b^d} \geq 0$ for $j = K$. For $j < K$, (5.1) gives $\lambda \sqrt{\alpha'_j} \leq \beta \sqrt{\log(2b)\, g(0)/2} \leq \kappa_0 \sqrt{d \log b}$ with $\kappa_0 := \beta \sqrt{g(0)/d}$. Lemma 3.5 then yields $\psi_{b^d}(\lambda \sqrt{\alpha'_j}) \geq \mathsf{R}(\lambda \sqrt{\alpha'_j}; d \log b) - d \log b\, \varrho_{b^d}(\kappa_0)$. By monotonicity of R, (5.1) and (4.6),

$$\mathsf{R}\big(\lambda \sqrt{\alpha'_j}; d \log b\big) \geq \mathsf{R}\Big(\beta \sqrt{(1-\eta)\log(2b) g(z_{j+1})/2};\, d \log b\Big) = \log b\, G_d\Big(\beta \sqrt{(1-\eta)\zeta\, g(z_{j+1})}\Big),$$

where $\zeta := \log(2b)/\log b$. Since $y \mapsto G_d(\beta\sqrt{(1-\eta)\zeta g(y)})$ is non-increasing,

$$\varepsilon \sum_{j=1}^{K-1} \log b\, G_d\big(\beta\sqrt{(1-\eta)\zeta g(z_{j+1})}\big) \geq \frac{1}{\zeta}\int_{z_2}^{z_{K+1}} G_d\big(\beta\sqrt{(1-\eta)\zeta g(y)}\big)dy \geq \frac{1}{\zeta}\int_{z_2}^{Y} G_d\big(\beta\sqrt{(1-\eta)\zeta g(y)}\big)dy,$$

for ε so small that $z_2 \leq Y$. Collecting terms, we use $\varepsilon K \leq (Y+h')/\log(2b)$ for the term $-\varepsilon d K \log 2$, and $\varrho_{b^d}(\kappa_0) \geq 0$ (since $\mathsf{R}(0;\ell) = \psi_n(0) = 0$) for the term $-\varepsilon(K-1)d\log b\,\varrho_{b^d}(\kappa_0) \geq -d(Y+h')\varrho_{b^d}(\kappa_0)$. Letting $\varepsilon \downarrow 0$ (so $\eta \to 0$, $h' \to 0$, $\varepsilon \log q_\varepsilon \to 0$ and hence $z_2 \to 0$), dominated convergence gives

$$\liminf_{\varepsilon\downarrow 0} \varepsilon\,\mathbb{E}\log \int_{\mathsf{T}} e^{\beta X^\varepsilon/\sqrt{\varepsilon}} \geq \frac{1}{\zeta}\int_0^Y G_d\big(\beta\sqrt{\zeta g}\big) - \frac{dY\log 2}{\log(2b)} - dY\,\varrho_{b^d}(\kappa_0) - \beta\sqrt{2dY\Phi(Y)}.$$

Let $b \to \infty$: then $\zeta \to 1$, so $\frac{1}{\zeta}\int_0^Y G_d(\beta\sqrt{\zeta g}) \to \int_0^Y G_d(\beta\sqrt{g})$ by bounded convergence (as $1 \leq \zeta \leq 2$ and $G_d(\beta\sqrt{\zeta g}) \leq G_d(\beta\sqrt{2g(0)})$), and $\varrho_{b^d}(\kappa_0) \to 0$ by Lemma 3.5, since κ_0 does not depend on b. This gives the lower bound $\int_0^Y G_d(\beta\sqrt{g}) - \beta\sqrt{2dY\Phi(Y)}$. Finally let $Y \to \infty$ and use (2.3). □

6. Proofs of the main theorems

6.1. **Universality.**

Proof of Theorem 2.7. Parts (i) and (ii) follow by combining Propositions 4.5 and 5.5. For (iii), fix $t_0 \in \mathsf{T}$ and put $\widetilde{X}^\varepsilon := X^\varepsilon - X^\varepsilon_{t_0}$. By (2.6), $\sup_t \operatorname{Var}\widetilde{X}^\varepsilon_t \leq C_0$. By Borell–TIS (Lemma 3.3(b)), applied on a countable dense subset of T (by path continuity the suprema coincide), $\sup\widetilde{X}^\varepsilon - \mathbb{E}\sup\widetilde{X}^\varepsilon$ has sub-Gaussian tails with variance proxy C_0, uniformly in ε. Since $\mathbb{E}\sup\widetilde{X}^\varepsilon = \mathbb{E}\sup X^\varepsilon$,

$$\sqrt{\varepsilon}\sup_{\mathsf{T}} X^\varepsilon - \sqrt{\varepsilon}\,\mathbb{E}\sup_{\mathsf{T}} X^\varepsilon = \sqrt{\varepsilon}\,X^\varepsilon_{t_0} + \sqrt{\varepsilon}\big(\sup\widetilde{X}^\varepsilon - \mathbb{E}\sup\widetilde{X}^\varepsilon\big) \longrightarrow 0 \quad \text{in } L^p,$$

because $\operatorname{Var}(\sqrt{\varepsilon}X^\varepsilon_{t_0}) = o(1)$. For the free energy let $\lambda = \beta/\sqrt{\varepsilon}$ and write $\log\int_{\mathsf{T}} e^{\lambda X^\varepsilon} = \lambda X^\varepsilon_{t_0} + f$ with $f := \log\int_{\mathsf{T}} e^{\lambda\widetilde{X}^\varepsilon}$. By Lemma 3.3(c), $\mathbb{P}(|f - \mathbb{E}f| > u) \leq 2e^{-u^2/(2\lambda^2 C_0)}$, so the fluctuations of εf are $O(\beta\sqrt{C_0\varepsilon})$ in every L^p. Moreover $\varepsilon\lambda X^\varepsilon_{t_0} = \beta\sqrt{\varepsilon}X^\varepsilon_{t_0} \to 0$ in L^p. Since $\mathbb{E}f = \mathbb{E}\log\int e^{\lambda X^\varepsilon}$, the claim follows from (ii). □

6.2. **Structure of the limiting free energy.**

Proof of Proposition 2.9. (a) For each y, $\beta \mapsto G_d(\beta\sqrt{g(y)})$ is convex, non-decreasing and C^1, with derivative $\sqrt{g}\,G_d'(\beta\sqrt{g}) = \min(\beta g/2, \sqrt{dg}) \leq \sqrt{d}\sqrt{g}$, which is integrable. Hence F is convex, non-decreasing and C^1, with the stated derivative, by dominated convergence. If $\beta \leq \beta_c$ then $\beta\sqrt{g(y)} \leq \beta\sqrt{g(0)} \leq 2\sqrt{d}$ for all y, so $F(\beta) = \frac{\beta^2}{4}\int_0^\infty g = \frac{\beta^2}{4}\Phi(0)$. Finally $G_d(\beta\sqrt{g})/\beta \to \sqrt{dg}$ pointwise and $G_d(z) \leq \sqrt{d}z$, so $F(\beta)/\beta \to \sqrt{d}\,I_\Phi$.

(b) If Φ is strictly convex, g is continuous, strictly decreasing and tends to 0. For $\beta > \beta_c$ the equation $\beta\sqrt{g(y)} = 2\sqrt{d}$ has a unique solution $y_*(\beta) > 0$, which is continuous and increasing in β, with $y_*(\beta) \downarrow 0$ as $\beta \downarrow \beta_c$. Then

$$F'(\beta) = \int_0^{y_*(\beta)} \sqrt{dg}\,dy + \frac{\beta}{2}\int_{y_*(\beta)}^\infty g\,dy = \int_0^{y_*(\beta)} \sqrt{dg}\,dy + \frac{\beta}{2}\Phi(y_*(\beta)).$$

For fixed y, the map $\beta \mapsto \min(\beta g(y)/2, \sqrt{dg(y)})$ is Lipschitz with almost-everywhere derivative $\frac{g(y)}{2}\mathbf{1}\{\beta\sqrt{g(y)} < 2\sqrt{d}\}$. By Fubini, $F'(\beta_2) - F'(\beta_1) = \int_{\beta_1}^{\beta_2} \frac{1}{2}\Phi(y_*(\beta))\,d\beta$ for $\beta_c < \beta_1 < \beta_2$, and by continuity of F' also for $\beta_1 = \beta_c$, with $y_*(\beta_c) := 0$. Since $\beta \mapsto \Phi(y_*(\beta))$ is continuous and tends to $\Phi(0)$ as $\beta \downarrow \beta_c$, F'' exists on $(0,\infty)$, equals $\frac{1}{2}\Phi(0)$ on $(0,\beta_c]$ and $\frac{1}{2}\Phi(y_*(\beta))$ on (β_c,∞), and is continuous.

(c) If $\Phi \in C^2([0,y_0))$ with $\Phi''(0) > 0$, then $g' = -\Phi'' < 0$ on some interval $[0,y_1)$, and g' is continuous at 0 from the right. The implicit function theorem applied to $\beta\sqrt{g(y)} = 2\sqrt{d}$ gives $y_* \in C^1$ on $(\beta_c, \beta_c+\theta)$ for some $\theta > 0$, with $y_*'(\beta) = -2g(y_*)/(\beta g'(y_*))$. Therefore

$$F'''(\beta) = \tfrac{1}{2}\Phi'(y_*)y_*' = \frac{g(y_*)^2}{\beta\, g'(y_*)} \xrightarrow[\beta\downarrow\beta_c]{} \frac{g(0)^2}{\beta_c g'(0)} = -\frac{\Phi'(0)^2}{\beta_c \Phi''(0)},$$

whereas $F''' = 0$ on $(0,\beta_c)$. □

6.3. **Fractional Brownian motion.**

Proof of Theorem 2.1(iv). Apply Lemma 3.3(b) to B^H on the countable set $D := [0,1]\cap\mathbb{Q}$. By path continuity $\sup_D B^H = \sup_{[0,1]} B^H$, so $\mathbb{E}\sup_D B^H = M_H < \infty$ by part (i), and $\sup_t \operatorname{Var} B_t^H = 1$, which gives the tail bound. Since $\sqrt{H}M_H \to \sqrt{2}$ by part (iii), $\sqrt{H}\sup B^H - \sqrt{2} = \sqrt{H}(\sup B^H - M_H) + (\sqrt{H}M_H - \sqrt{2}) \to 0$ in L^p. □

Proof of Theorem 2.2. fBm on $[0,1]$ with $\varepsilon = H$ satisfies Assumption 2.5 with $d=1$, $\Phi(y) = e^{-2y}$, $\eta \equiv 0$, $r_\varepsilon = q_\varepsilon = 1$, $Q_\varepsilon = [0,1]$ and $C_0 = 1$, because $\mathbb{E}(B_t^H - B_s^H)^2 = |t-s|^{2H} = \Phi(H\log(1/|t-s|))$ exactly. Moreover $\operatorname{Var} B_t^H \le 1$. Φ is a strictly convex profile with $g(y) = 2e^{-2y}$, $I_\Phi = \sqrt{2}$ and $\beta_c = 2/\sqrt{g(0)} = \sqrt{2}$. By Theorem 2.7, $H\log Z_H(\beta) \to F_{\Phi,1}(\beta)$ in L^p. If $\beta \le \sqrt{2}$, Proposition 2.9(a) gives $F(\beta) = \beta^2/4$. If $\beta > \sqrt{2}$, then $y_* = \log(\beta/\sqrt{2})$ and

$$F(\beta) = \int_0^{y_*} \big(\sqrt{2}\beta e^{-y} - 1\big)dy + \int_{y_*}^{\infty} \frac{\beta^2}{2}e^{-2y}dy = \sqrt{2}\beta\Big(1 - \frac{\sqrt{2}}{\beta}\Big) - \log\frac{\beta}{\sqrt{2}} + \frac{1}{2} = \sqrt{2}\beta - \frac{3}{2} - \log\frac{\beta}{\sqrt{2}}.$$

The derivatives $F'' = \frac{1}{2}$ on $(0,\sqrt{2})$ and $F''(\beta) = \beta^{-2}$, $F'''(\beta) = -2\beta^{-3}$ on $(\sqrt{2},\infty)$ are immediate, and $F'''(\sqrt{2}+) = -1/\sqrt{2}$, in accordance with Proposition 2.9(c). □

6.4. **Examples.**

Proof of Corollary 2.11(a). Since $|u| \le \|u\|_2 \le \sqrt{d}\,|u|$, we have $\Phi(H\log(1/|t-s|)) \le \|t-s\|_2^{2H} \le d^H\Phi(H\log(1/|t-s|))$ with $\Phi(y) = e^{-2y}$. So Assumption 2.5 holds with $\varepsilon = H \in (0,\varepsilon_0)$, where $\varepsilon_0 := \log 2/\log(2d) \le 1$ ensures $\eta < 1$, with $\eta(H) = d^H - 1$, $r_\varepsilon = q_\varepsilon = 1$, $Q_\varepsilon = \mathsf{T}$ and $C_0 = d^H$. The field has continuous paths and $\operatorname{Var} X_t^H \le d^H$. Here $\sqrt{d}\,I_\Phi = \sqrt{2d}$, $\beta_c = \sqrt{2d}$, and for $\beta > \sqrt{2d}$, with $y_* = \log(\beta/\sqrt{2d})$,

$$F_d(\beta) = \int_0^{y_*} \big(\sqrt{2d}\,\beta e^{-y} - d\big)dy + \frac{\beta^2}{4}e^{-2y_*} = \sqrt{2d}\,\beta - 2d - d\,y_* + \frac{d}{2} = \sqrt{2d}\,\beta - \frac{3}{2}d - d\log\frac{\beta}{\sqrt{2d}}.$$ □

For the Riemann–Liouville process we need an elementary estimate.

Lemma 6.1. *Let $H \in (0,\frac{1}{4}]$ and $0 \le s < t \le 1$, $\delta := t-s$. Then*

$$\mathbb{E}(R_t^H - R_s^H)^2 = \delta^{2H}\big(1 + 2HJ_H(s/\delta)\big), \qquad J_H(R) := \int_0^R \big(w^{H-1/2} - (1+w)^{H-1/2}\big)^2 dw,$$

and $\delta^{2H} \le \mathbb{E}(R_t^H - R_s^H)^2 \le (2 + H/3)\,\delta^{2H}$. If moreover $s \ge \delta$, then $\mathbb{E}(R_t^H - R_s^H)^2 \ge (2 - 16H)\,\delta^{2H}$.

Proof. By the Itô isometry, $\mathbb{E}(R_t^H - R_s^H)^2 = 2H\int_s^t (t-u)^{2H-1}du + 2H\int_0^s \big((t-u)^{H-1/2} - (s-u)^{H-1/2}\big)^2 du$. The substitution $s-u = \delta w$ gives the formula. For $w \in (0,1)$, $0 \le w^{H-1/2} - (1+w)^{H-1/2} \le w^{H-1/2}$. For $w \ge 1$ the mean value theorem gives $0 \le w^{H-1/2} - (1+w)^{H-1/2} \le (\frac12 - H)w^{H-3/2}$. Hence $J_H(\infty) \le \frac{1}{2H} + \frac{(1/2-H)^2}{2-2H} \le \frac{1}{2H} + \frac16$, which proves the upper bound. For the last claim note that $J_H(R) \ge J_H(1)$ for $R \ge 1$, and that the integrand equals $w^{2H-1}\big(1 - (w/(1+w))^{1/2-H}\big)^2 \ge w^{2H-1}(1 - 2w^{1/4})$ for $w \in (0,1)$, because $(w/(1+w))^{1/2-H} \le w^{1/4}$. Thus $J_H(1) \ge \frac{1}{2H} - \frac{2}{2H+1/4} \ge \frac{1}{2H} - 8$, and $\mathbb{E}(R_t^H - R_s^H)^2 \ge \delta^{2H}\big(1 + 2H(\frac{1}{2H} - 8)\big) = (2 - 16H)\delta^{2H}$. □

Proof of Corollary 2.11(b)–(d). (b) By Lemma 6.1, Assumption 2.5 holds with $\varepsilon = H \in (0, \frac{1}{16})$, $\Phi(y) = 2e^{-2y}$, $\eta = 8H$, $r_\varepsilon = 1$, $C_0 = 2\Phi(0)$ and $Q_\varepsilon = [\frac12, 1]$, so $q_\varepsilon = \frac12$: for $s,t \in Q_\varepsilon$ we have $\min(s,t) \ge \frac12 \ge |t-s|$. The paths are continuous and $\operatorname{Var} R_t^H = t^{2H} \le 1$. Here $g = 4e^{-2y}$, $I_\Phi = 2$ and $\beta_c = 1$. For $\beta > 1$, $y_* = \log\beta$ and $F(\beta) = \int_0^{\log\beta}(2\beta e^{-y} - 1)dy + \beta^2 e^{-2\log\beta}/2 = 2\beta - \frac32 - \log\beta$.

(c) Let $V(r) := \int r^{2\varepsilon h}\mu(dh)$. Since $\varepsilon h_+ < 1$, the kernel $\frac12(V(s) + V(t) - V(|t-s|))$ is a mixture of fBm covariances, hence positive semi-definite, and X^ε exists; its law is determined by V. For $r \le 1$ we have $V(r) \le \mu(\mathbb{R})r^{2\varepsilon h_-}$, so X^ε has a continuous version by Kolmogorov's criterion, and $\operatorname{Var} X_t^\varepsilon = V(t) \le \mu(\mathbb{R})$. Moreover $\mathbb{E}(X_t^\varepsilon - X_s^\varepsilon)^2 = \Phi_\mu(\varepsilon\log(1/|t-s|))$ exactly. Differentiation under the integral shows that Φ_μ is C^2 with $\Phi'_\mu(y) = -\int 2he^{-2hy}\mu(dh) < 0$ and $\Phi''_\mu \ge 0$; also $\Phi_\mu(y) \to 0$, and $g(y) \le 2h_+\mu(\mathbb{R})e^{-2h_- y}$, so $I_\Phi < \infty$. Thus Φ_μ is a profile, and Assumption 2.5 holds with $d = 1$, $\eta \equiv 0$, $r_\varepsilon = q_\varepsilon = 1$, $Q_\varepsilon = [0,1]$ and $C_0 = \mu(\mathbb{R})$. The claims follow from Theorem 2.7. For $\mu = \sum_k c_k^2\delta_{h_k}$ the process $\sum_k c_k B^{(k)}$ has the same variogram and $X_0 = 0$, hence the same law.

(d) For $2H \le 2$, the function $\tau \mapsto \exp(-\|\tau\|_2^{2H})$ is the characteristic function of a rotationally invariant $2H$-stable law, so it is positive definite and X^H exists. Let $\Psi(x) := 2(1 - e^{-x})$, so that $\mathsf{d}(s,t)^2 = \Psi(\|t-s\|_2^{2H})$ and $\Phi(y) = \Psi(e^{-2y})$. Since Ψ is increasing and concave with $\Psi(0) = 0$, we have $\Psi(x) \le \Psi(cx) \le c\Psi(x)$ for $c \ge 1$. With $|u| \le \|u\|_2 \le \sqrt{d}\,|u|$ this gives

$$\Phi\big(H\log(1/|t-s|)\big) \le \mathsf{d}(s,t)^2 \le d^H\,\Phi\big(H\log(1/|t-s|)\big).$$

The function $g(y) = 4we^{-w}$, $w = e^{-2y}$, is decreasing in $y \ge 0$ because $w \mapsto we^{-w}$ is increasing on $[0,1]$. So Φ is a strictly convex profile, and Assumption 2.5 holds with $\varepsilon = H < \log 2/\log(2d)$, $\eta = d^H - 1$, $r_\varepsilon = q_\varepsilon = 1$, $Q_\varepsilon = \mathsf{T}$ and $C_0 = 2$. The paths are continuous by Kolmogorov's criterion, since $\mathsf{d}(s,t)^2 \le 2\|t-s\|_2^{2H}$, and $\operatorname{Var} X_t^H = 1$. The substitution $u = e^{-y}$ gives $I_\Phi = \int_0^\infty 2e^{-y}e^{-e^{-2y}/2}dy = 2\int_0^1 e^{-u^2/2}du$, and $g(0) = 4/e$ gives $\beta_c = \sqrt{e\,d}$. Finally, $g'(0) = 0$ and $g''(0) = -16/e$, so $\Phi''(0) = 0$ and $\Phi'''(0) = 16/e > 0$. □

6.5. **Pickands' constants.**

Proof of Corollary 2.16. By Shao's inequality and Theorem 2.1(iii), $\sqrt{2}\,eHM_H \le 2e\sqrt{H}\big(1 + H(2+\log(1/H))/\sqrt{2}\big)$. Raising to the power $1/H$ and using $(1+x)^{1/H} \le e^{x/H}$ gives the bound. Taking logarithms, $H\log\mathcal{H}_{2H} - \frac12\log H \le \log(2e) + H(2+\log(1/H))/\sqrt{2} \to \log(2e)$. □

7. Maxima over grids

7.1. **Upper bounds.**

Proof of Proposition 2.12. Let $(\zeta_t)_{t\in\Gamma}$ be i.i.d. $N(0,\frac12)$. For $s \ne t$ in Γ, $\mathbb{E}(\zeta_t - \zeta_s)^2 = 1 \ge |t-s|^{2H}$, so Corollary 3.2 gives $\mathbb{E}\max_\Gamma B^H \le \mathbb{E}\max_\Gamma \zeta = \mu_n/\sqrt{2} \le \sqrt{\log n}$. □

Proposition 7.1. *Let $H \downarrow 0$ and $H \log n \to \kappa \in [0,\infty]$. Then* $\limsup \sqrt{H}\, \mathbb{E} \max_{\Gamma_n} B^H \le \Lambda(\kappa)$.

Proof. If $\kappa \le \frac{1}{2}$ this follows from Proposition 2.12, and if $\kappa = \infty$ from $\mathbb{E}\max_{\Gamma_n} B^H \le M_H$ and Theorem 2.1. Let $\kappa \in (\frac{1}{2}, \infty)$, and fix $b = 2$ and $\delta \in (0,1)$. Put $L_* := \log b/(b^{2H}-1)$ and let K be the largest integer with $\log n - K \log b \ge L_*$; for H small this is well defined (since $\kappa > \frac{1}{2}$ and $HL_* \to \frac{1}{2}$) and $K \to \infty$. Put $N_f := \lceil n b^{-K} \rceil + 1$, so that $L_* \le \log N_f \le L_* + \log b + \log 3$.

Every interval of $\mathcal{Q}_K$ (the b-adic intervals of length b^{-K}) contains at most N_f points of Γ_n. Label the points of $\Gamma_n \cap Q$ injectively by $\{1,\dots,N_f\}$ and define, for $t \in \Gamma_n \cap Q$,

$$Y_t := S_Q + \sqrt{\nu}\, \xi'_{Q,\mathrm{lab}(t)}, \qquad \nu := \tfrac{1}{2} b^{-2HK},$$

where S is the GREM of Proposition 4.4 with $\varphi(r) = r^{2H}$, i.e. $\alpha_j = \frac{1}{2}(1-b^{-2H}) b^{-2H(j-1)}$, and the ξ' are independent standard Gaussians. If s,t lie in the same Q, then $|t-s|^{2H} \le b^{-2HK} = 2\nu = \mathbb{E}(Y_t - Y_s)^2$. Otherwise s,t lie in different cubes whose deepest common ancestor has some depth $j-1$; then s,t lie in one cube of $\mathcal{Q}_{j-1}$, so $|t-s| \le b^{-j+1}$, and the computation of Proposition 4.4 gives $\mathbb{E}(Y_t - Y_s)^2 = b^{-2H(j-1)} \ge |t-s|^{2H}$. Hence $\mathbb{E}\max_{\Gamma_n} B^H \le \mathbb{E}\max_{\Gamma_n} Y \le \mathbb{E}\max S^+$, where S^+ is the GREM with parameters $(b,\alpha_1),\dots,(b,\alpha_K),(N_f,\nu)$. This GREM has a concave profile: the θ_j increase for $j \le K$, and $\theta_{K+1}^2 = 2\log N_f/\nu \ge 2\log b/\alpha_K = \theta_K^2$ is equivalent to $\log N_f \ge \log b \cdot \nu/\alpha_K = L_*$. By Corollary 4.2,

$$\mathbb{E}\max_{\Gamma_n} B^H \le (1+\delta)\Big[\sqrt{\log b \coth(\tfrac{1}{2}H\log b)}\,\big(1-b^{-HK}\big) + b^{-HK}\sqrt{\log N_f}\Big] + \frac{1+\log^+ P_\delta}{c_1},$$

with $P_\delta \le (b^{2\delta}-1)^{-1}+1$ (the last summand, from level $K+1$, is at most 1) and $c_1 \ge \sqrt{2/H}$. As $H \downarrow 0$: $HL_* \to \frac{1}{2}$, hence $H\log N_f \to \frac{1}{2}$ and $HK\log b \to \kappa - \frac{1}{2}$. Moreover $\sqrt{H}\sqrt{\log b \coth(\frac{1}{2}H\log b)} \to \sqrt{2}$. Therefore $\limsup \sqrt{H}\,\mathbb{E}\max_{\Gamma_n} B^H \le (1+\delta)\big[\sqrt{2}(1-e^{-(\kappa-1/2)}) + e^{-(\kappa-1/2)}/\sqrt{2}\big] = (1+\delta)\Lambda(\kappa)$, and $\delta \downarrow 0$ concludes. □

7.2. **A second moment estimate for sub-chord GREMs.** For lower bounds in the regime $\kappa \le \frac{1}{2}$, and on the last block of scales when $\kappa > \frac{1}{2}$, the hierarchical structure is *sub-critical*: the profile lies below its chord, and the maximum is that of a random energy model. Greedy bounds are then far from sharp, and we use a truncated second moment argument. For a fixed number of levels, the fact that a GREM whose profile lies below its chord has the maximum of a random energy model is contained in [20, Thm 1.1], and the analogue for branching Brownian motion is the weak-correlation regime of variable-speed BBM [19]. Lemma 7.2 is a quantitative version for a growing number of levels. Its proof is the truncated second moment method in the form of Kistler's multiscale refinement [54]; cf. [11, 29, 5, 4, 71].

Lemma 7.2. *Let $b \ge 2$ and $K \ge 2$ be integers, and consider the GREM with parameters $(b,\alpha_j)_{j\le K}$, where all $\alpha_j > 0$. Let $A_j := \sum_{i\le j}\alpha_i$, $A := A_K$ and $\mathcal{E} := K\log b$, and assume the sub-chord condition $A_j \le \frac{j}{K}A$ for $1 \le j \le K$. Let $\varpi \in (0,\frac{1}{2}]$, $c_0 := \frac{7}{8}e^{-3/2}(2\pi)^{-1/2}$, and assume*

(S1) $c_0 e^{\varpi\mathcal{E}} \ge \sqrt{2\mathcal{E}}$ *and* $2(1-\varpi)^2\mathcal{E} \ge 1$;
(S2) $b^{\varpi/2} \ge 1 + 2e\, c_0^{-2}\mathcal{E}\,(2K)^{4/\varpi}$;
(S3) $\varpi\mathcal{E} \ge 2\sqrt{2}\sqrt{\mathcal{E}\log(2K)} + \log\big(4e^2 c_0^{-2}\mathcal{E}K\big)$.

Then $\mathbb{E}\max_{v\in\mathcal{T}_K} S_v \ge (1-\varpi)\sqrt{2A\mathcal{E}} - 3\sqrt{A}$.

Proof. Let $x := (1-\varpi)\sqrt{2A\mathcal{E}}$, so that $x^2/(2A) = (1-\varpi)^2\mathcal{E}$ and, by (S1), $x^2 \geq A$. For a leaf v write $S_j = S_{v_j}$, and let

$$w_j := \frac{A_j}{x} + 2\sqrt{A_j \log(2K)}, \quad N := \#\Big\{v \in \mathcal{T}_K : S_K \in \big[x, x + \tfrac{A}{x}\big],\ S_j \leq \tfrac{A_j}{A}x + w_j\ \forall j < K\Big\}.$$

First moment. Fix a leaf and write $S_j = \frac{A_j}{A}S_K + \mathcal{R}_j$. Here $\mathcal{R}_j$ is Gaussian, independent of S_K, with $\operatorname{Var}\mathcal{R}_j \leq A_j$. On $\{S_K \in [x, x + A/x]\}$ the barrier condition holds whenever $\mathcal{R}_j \leq 2\sqrt{A_j \log(2K)}$ for all $j < K$, which fails with probability at most $(K-1)(2K)^{-2} \leq \frac{1}{4K} \leq \frac{1}{8}$ (here $K \geq 2$ is used). Since $(x + A/x)^2 \leq x^2 + 3A$, we get $\mathbb{P}(S_K \in [x, x + A/x]) \geq \frac{\sqrt{A}}{x\sqrt{2\pi}} e^{-3/2} e^{-x^2/(2A)}$. Hence

$$\mathbb{E}N \geq c_0 \frac{\sqrt{A}}{x}\, e^{\mathcal{E} - x^2/(2A)} \geq c_0 \frac{e^{\varpi\mathcal{E}}}{\sqrt{2\mathcal{E}}} \geq 1, \tag{7.1}$$

using $\mathcal{E} - x^2/(2A) = (2\varpi - \varpi^2)\mathcal{E} \geq \varpi\mathcal{E}$, $x/\sqrt{A} \leq \sqrt{2\mathcal{E}}$ and (S1).

Second moment. Write $\mathbb{E}N^2 = \sum_{j=0}^{K} \Sigma_j$, where Σ_j is the contribution of ordered pairs of leaves whose deepest common ancestor has depth j. Then $\Sigma_K = \mathbb{E}N$, and for $j < K$ there are at most b^{2K-j} such pairs. For such a pair let $U = S_j$ (with $U = 0$ if $j = 0$), and write the two values of S_K as $U + V$ and $U + V'$, with V, V' i.i.d. $N(0, A - A_j)$ independent of U. Let $\ell_j := \frac{A_j}{A}x + w_j$. Put $\tau_j := b^{2K-j} P_j/(\mathbb{E}N)^2$, where P_j bounds the probability that both leaves are counted, and recall $(\mathbb{E}N)^2 \geq c_0^2 (A/x^2) e^{2\mathcal{E} - x^2/A}$.

$j = 0$. $P_0 \leq \mathbb{P}(S_K \geq x)^2 \leq \frac{A}{2\pi x^2} e^{-x^2/A}$, so $\tau_0 \leq (2\pi c_0^2)^{-1} < 26.3$.

$1 \leq j < K$, $x \geq \ell_j$. Using $\mathbb{P}(V \geq z) \leq e^{-z^2/(2(A-A_j))}$ for $z \geq 0$,

$$P_j := \mathbb{E}\Big[\mathbf{1}\{U \leq \ell_j\}\, \mathbb{P}(V \geq x - U \mid U)^2\Big] \leq \int_{-\infty}^{\ell_j} \frac{e^{f(u)}}{\sqrt{2\pi A_j}}\, du, \qquad f(u) := -\frac{u^2}{2A_j} - \frac{(x-u)^2}{A - A_j}.$$

Completing the square, $f(u) = f(u^*) - (u - u^*)^2/(2s^2)$ with $u^* = 2A_j x/(A + A_j)$ and $s^2 = A_j(A - A_j)/(A + A_j) \leq A_j$. If $\ell_j \geq u^*$, the integral is at most $(s/\sqrt{A_j}) e^{f(u^*)} \leq e^{f(u^*)}$. If $\ell_j < u^*$, then $(u - u^*)^2 \geq (u - \ell_j)^2 + (\ell_j - u^*)^2$ for $u \leq \ell_j$, so the integral is at most $(s/\sqrt{A_j}) e^{f(\ell_j)} \leq e^{f(\ell_j)}$. In both cases it is at most $\exp(\max_{u \leq \ell_j} f(u))$. At $u_0 := \frac{A_j}{A}x$ one computes $f(u_0) = -\frac{x^2}{A}(1 - \frac{A_j}{2A})$ and $f'(u_0) = x/A > 0$. By concavity, $\max_{u \leq \ell_j} f \leq f(u_0) + x w_j/A$. Thus

$$\tau_j \leq \frac{x^2}{c_0^2 A} \exp\Big(-j \log b + \frac{x^2 A_j}{2A^2} + \frac{x w_j}{A}\Big).$$

By the sub-chord condition, $\frac{x^2 A_j}{2A^2} = (1-\varpi)^2 \mathcal{E} \frac{A_j}{A} \leq (1-\varpi)^2 j \log b$. Moreover $\frac{x w_j}{A} = \frac{A_j}{A} + \frac{2x}{A}\sqrt{A_j \log 2K} \leq 1 + 2\sqrt{2j \log b \log(2K)} \leq 1 + \frac{\varpi}{2} j \log b + \frac{4}{\varpi}\log(2K)$. Using $x^2/A \leq 2\mathcal{E}$ we obtain $\tau_j \leq 2e\, c_0^{-2} \mathcal{E}\, (2K)^{4/\varpi} b^{-\varpi j/2}$.

$1 \leq j < K$, $x < \ell_j$. Without the barrier, $P_j \leq \mathbb{P}(U \geq x) + \mathbb{E}[e^{-(x-U)^2/(A-A_j)}; U < x] \leq e^{-x^2/(2A_j)} + e^{f(u^*)} \leq 2e^{-x^2/(A+A_j)}$. Since $\frac{x^2}{A} - \frac{x^2}{A+A_j} \leq \frac{x^2}{2A} = (1-\varpi)^2\mathcal{E}$,

$$\tau_j \leq \frac{4\mathcal{E}}{c_0^2} \exp\big(-j \log b + (1-\varpi)^2 \mathcal{E}\big).$$

In this case $x(1 - A_j/A) < w_j \leq A/x + 2\sqrt{A \log(2K)}$, and $1 - A_j/A \geq (K-j)/K$. This gives $(K-j)\log b < \frac{KA \log b}{x^2} + \frac{2K \log b}{x}\sqrt{A \log 2K} \leq 2 + 2\sqrt{2}\sqrt{\mathcal{E}\log(2K)}$, because $KA \log b/x^2 =$

$1/(2(1-\varpi)^2) \le 2$. Hence $-j\log b + (1-\varpi)^2\mathcal{E} \le -\varpi\mathcal{E} + 2 + 2\sqrt{2}\sqrt{\mathcal{E}\log(2K)}$, and there are at most K such indices j.

Conclusion. By (S2) and (S3), the sums of τ_j over the last two cases are each at most 1. With (7.1), $\mathbb{E}N^2/(\mathbb{E}N)^2 \le 1/\mathbb{E}N + 26.3 + 2 < 30$. By the Paley–Zygmund inequality, $\mathbb{P}(\max_{\mathcal{T}_K} S \ge x) \ge \mathbb{P}(N>0) \ge (\mathbb{E}N)^2/\mathbb{E}N^2 > \frac{1}{30}$. Since $\max S$ is $\sqrt{A}$-Lipschitz in the underlying Gaussian vector, (3.2) yields $\mathbb{E}\max S \ge x - \sqrt{A}\sqrt{2\log 30} \ge x - 3\sqrt{A}$. □

7.3. **Lower bounds.**

Proposition 7.3. *For $H, \Delta \in (0,1)$ let $\vartheta(H,\Delta) := \inf_\Gamma \mathbb{E}\max_{t\in\Gamma} B_t^H$, where the infimum is over all finite $\Gamma \subset [0,1]$ such that every closed subinterval of $[0,1]$ of length 2Δ meets Γ. If $H \downarrow 0$ and $\Delta \downarrow 0$ with $H\log(1/\Delta) \to \kappa' \in (0,\infty]$, then*

$$\liminf \sqrt{H}\,\vartheta(H,\Delta) \ge \sqrt{\min(\kappa', 1/2)}.$$

Proof. Let $L_H := \log(1/H)$, $b := \lceil e^{L_H^3} \rceil$, $\varpi := L_H^{-1/2}$, and

$$K := \min\Big(\Big\lfloor \frac{\log(1/(2\Delta))}{\log(2b)} \Big\rfloor,\ \Big\lfloor \frac{1}{2H\log(2b)} \Big\rfloor \Big).$$

Since $H\log(2b) \asymp HL_H^3 \to 0$, we have $K \to \infty$ and $HK\log(2b) \to \min(\kappa', \frac12)$. Apply the Cantor construction in $[0,1]$ with parameter b. Every cell of $\mathcal{C}_K$ has length $r_K = (2b)^{-K} \ge 2\Delta$, hence contains a point $t_C \in \Gamma$. By Lemma 5.2 with $\varphi(r) = r^{2H}$, $\mathbb{E}\max_\Gamma B^H \ge \mathbb{E}\max S'$, where S' is the GREM with parameters (b, α_j'), $\alpha_j' = \frac12(r_j^{2H} - r_{j+1}^{2H})$ for $j < K$ and $\alpha_K' = \frac12 r_K^{2H}$. All α_j' are positive. Moreover $A = \frac12 r_1^{2H} = \frac12 (2b)^{-2H}$ and, for $j < K$, $A_j/A = 1 - (2b)^{-2Hj} \le 2Hj\log(2b) \le j/K$, because $2HK\log(2b) \le 1$. So the sub-chord condition holds.

With $\mathcal{E} = K\log b \asymp H^{-1}$, $\log K \asymp L_H$, $\log b \asymp L_H^3$ and $\varpi = L_H^{-1/2}$, conditions (S1)–(S3) hold for H small. Indeed, in (S1) $\varpi\mathcal{E} \gg \log\mathcal{E}$. In (S2), $\frac{\varpi}{2}\log b \asymp L_H^{5/2}$ dominates $\log\mathcal{E} + \frac{4}{\varpi}\log(2K) = O(L_H^{3/2})$. In (S3), $\varpi\mathcal{E} \asymp H^{-1}L_H^{-1/2}$ dominates $\sqrt{\mathcal{E}\log K} \asymp H^{-1/2}L_H^{1/2}$ and $\log(\mathcal{E}K) = O(L_H)$. Lemma 7.2 gives

$$\sqrt{H}\,\mathbb{E}\max_\Gamma B^H \ \ge\ (1-\varpi)\sqrt{2HA\mathcal{E}} - 3\sqrt{HA} \ =\ (1-\varpi)(2b)^{-H}\sqrt{HK\log b} \,-\, 3\sqrt{HA}.$$

The right-hand side does not depend on Γ. Since $(2b)^{-H} \to 1$, $HK\log b = HK\log(2b)\cdot\frac{\log b}{\log(2b)} \to \min(\kappa', \frac12)$ and $A \le \frac12$, the claim follows. □

Remark 7.4. The proof of Proposition 7.3 is purely asymptotic and gives no useful rate: the factor $1-\varpi$ with $\varpi = (\log(1/H))^{-1/2}$ converges to 1 very slowly. A rate could be obtained by optimising the choice of b and ϖ, which we have not attempted.

Lemma 7.5. *For $H \in (0,1)$ let $X_u := B_u^H - \int_0^1 B_v^H\,dv$, $u \in [0,1]$. Then*

$$\operatorname{Var} X_u = \frac{u^{2H+1} + (1-u)^{2H+1} - 1}{2H+1} + \frac{1}{2H+2} \le \frac{1}{2H+2} < \frac12.$$

Proof. With $\bar B = \int_0^1 B_v^H\,dv$ and $\operatorname{Cov}(B_u^H, B_v^H) = \frac12(u^{2H} + v^{2H} - |u-v|^{2H})$ one computes $\operatorname{Cov}(B_u^H, \bar B) = \frac12\big(u^{2H} + \frac{1-u^{2H+1}-(1-u)^{2H+1}}{2H+1}\big)$ and $\operatorname{Var}\bar B = \frac{1}{2H+1} - \frac12\iint |v-w|^{2H} = \frac{1}{2H+1} - \frac{1}{(2H+1)(2H+2)} = \frac{1}{2H+2}$. Then $\operatorname{Var} X_u = u^{2H} - 2\operatorname{Cov}(B_u^H, \bar B) + \operatorname{Var}\bar B$ gives the formula, and $u^{2H+1} + (1-u)^{2H+1} \le u + (1-u) = 1$ gives the bound. □

Proposition 7.6. *Let $H \downarrow 0$ and $H\log n \to \kappa \in [0,\infty]$. Then $\liminf \sqrt{H}\,\mathbb{E}\max_{\Gamma_n} B^H \ge \Lambda(\kappa)$.*

Proof. For $\kappa = 0$ there is nothing to prove. For $\kappa \in (0, \frac{1}{2}]$, apply Proposition 7.3 with $\Delta = 1/n$: every interval of length $2/n$ meets Γ_n. Let now $\kappa \in (\frac{1}{2}, \infty]$, and fix an integer $b \geq 2$ and $y_1 \in (0, \kappa - \frac{1}{2})$ (any $y_1 > 0$ if $\kappa = \infty$). Run the Cantor construction in $[0,1]$ up to depth $K_1 := \lfloor y_1/(H\log(2b)) \rfloor$, and let $\rho := r_{K_1} = (2b)^{-K_1}$, so that $\rho^H \to e^{-y_1}$. For $C \in \mathcal{C}_{K_1}$ with left endpoint a_C let $\Gamma^C := \Gamma_n \cap C$ and $U^C := \rho^{-1}(\Gamma^C - a_C) \subset [0,1]$. Let S' be the GREM on the Cantor tree of depth K_1 with $\alpha'_j := \frac{1}{2}(r_j^{2H} - r_{j+1}^{2H})$ for $j < K_1$ and $\alpha'_{K_1} := 0$. Let $(X^{(C)})_C$ be i.i.d. copies of the process X of Lemma 7.5, independent of S', and define

$$Y_t := S'_C + \rho^H X^{(C)}_{(t-a_C)/\rho}, \qquad t \in \Gamma^C,\ C \in \mathcal{C}_{K_1}.$$

Comparison. If $s, t \in \Gamma^C$, then $\mathbb{E}(Y_t - Y_s)^2 = \rho^{2H}|(t-s)/\rho|^{2H} = |t-s|^{2H}$, since X has the increments of B^H. If $t \in C$, $s \in C'$ with $C \neq C'$ whose deepest common ancestor has depth $j-1$, then $|t-s| \geq r_j$ by Lemma 5.1, and by Lemma 7.5

$$\mathbb{E}(Y_t - Y_s)^2 = 2\sum_{i=j}^{K_1} \alpha'_i + \rho^{2H}\big(\operatorname{Var} X_u + \operatorname{Var} X_{u'}\big) \leq \big(r_j^{2H} - \rho^{2H}\big) + \rho^{2H} = r_j^{2H} \leq |t-s|^{2H}.$$

Corollary 3.2 gives $\mathbb{E}\max_{\Gamma_n} B^H \geq \mathbb{E}\max_{\bigcup_C \Gamma^C} Y$.

Greedy choice. Let C^* be the leaf selected greedily from S' as in Proposition 5.3. Since C^* is $\sigma(S')$-measurable and the $X^{(C)}$ are i.i.d. and independent of S', and $\mathbb{E}\int_0^1 B^H = 0$,

$$\mathbb{E}\max_{\bigcup_C \Gamma^C} Y \geq \mathbb{E}S'_{C^*} + \rho^H\,\mathbb{E}\Big[\max_{u \in U^{C^*}} X_u^{(C^*)}\Big] \geq \mu_b \sum_{j<K_1} \sqrt{\alpha'_j} + \rho^H\,\vartheta\big(H, 1/(n\rho)\big).$$

For the last term, note that U^C has the covering property of Proposition 7.3 with $\Delta = 1/(n\rho)$. Indeed, C is a closed interval of length ρ and every subinterval of C of length $2/n$ meets Γ_n. Moreover $H\log(n\rho) \to \kappa - y_1 > \frac{1}{2}$. Hence $\liminf \sqrt{H}\,\vartheta(H, 1/(n\rho)) \geq 1/\sqrt{2}$.

For the first term, with $x = H\log(2b) \to 0$,

$$\sqrt{H}\mu_b \sum_{j<K_1} \sqrt{\alpha'_j} = \sqrt{H}\mu_b \sqrt{\frac{1-e^{-2x}}{2}} \sum_{j=1}^{K_1-1} e^{-xj} \longrightarrow \frac{\mu_b}{\sqrt{\log(2b)}}\big(1 - e^{-y_1}\big).$$

Altogether

$$\liminf \sqrt{H}\,\mathbb{E}\max_{\Gamma_n} B^H \geq \frac{\mu_b}{\sqrt{\log(2b)}}\big(1 - e^{-y_1}\big) + \frac{e^{-y_1}}{\sqrt{2}}.$$

Letting $b \to \infty$ (Lemma 3.4(c)) and then $y_1 \uparrow \kappa - \frac{1}{2}$ (or $y_1 \to \infty$ if $\kappa = \infty$) gives $\Lambda(\kappa)$. □

Proof of Theorem 2.13. Combine Propositions 7.1 and 7.6. □

8. Numerical illustration

Bounds for M_H. Table 1 evaluates the bounds of Theorem 2.1 and compares them with the best previously known bounds at the same H. For the upper bound we minimise (2.1) over $\delta > 0$ and over the integers $b \in \{2, \dots, 64\} \cup \{\lfloor 2^{k/2} \rceil : 13 \leq k \leq 40\}$. For the lower bound we maximise (2.2) over $b \in \{2^k : k \leq 400\}$, with μ_b computed by numerical quadrature, and over $b = \lceil e^L \rceil$ with $L \in [10^3, 10^9]$, with μ_b bounded below by Lemma 3.4(b). The entries are numerical evaluations of the bounds, not certified enclosures. Both normalised bounds converge to $\sqrt{2} \approx 1.4142$, the upper one much faster than the lower one, as predicted by Theorem 2.1(iii). Our upper bound improves on $1.695/\sqrt{H}$ [14] for all $H \leq 0.24$, and our lower bound improves on that of [7] for $H \leq 0.01$. For larger H the coupling bound of [7] is better; our lower bound is designed for the regime $H \downarrow 0$.

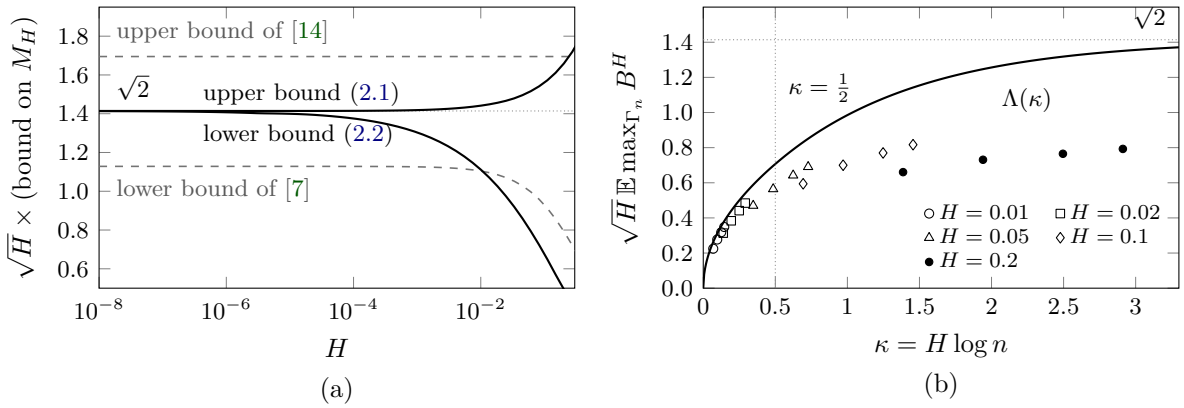


FIGURE 1. (a) The normalised bounds of Theorem 2.1, upper (2.1) and lower (2.2) (solid), and the previously best bounds, the upper bound of [14] and the lower bound of [7] (dashed), as functions of H. Our bounds converge to $\sqrt{2}$ (dotted). (b) The limit $\Lambda(\kappa)$ of Theorem 2.13 (solid) and the Monte Carlo estimates of $\sqrt{H}\,\mathbb{E}\max_{\Gamma_n} B^H$ for $n \in \{2^{10}, 2^{14}, 2^{18}, 2^{21}\}$; standard errors are smaller than the markers. For each n the ratio to $\Lambda(\kappa)$ increases as H decreases. The dotted lines mark $\kappa = \frac{1}{2}$ and the continuum value $\sqrt{2}$.

H	$\sqrt{2/H}$	lower (2.2)	lower [7]	upper (2.1)	upper [14]	$\sqrt{H}\cdot$[ours]
0.3	2.582	0.710	**1.276**	3.176	**3.095**	[0.389, 1.740]
0.1	4.472	2.133	**2.940**	**4.935**	5.360	[0.674, 1.561]
0.05	6.325	3.746	**4.554**	**6.717**	7.580	[0.838, 1.502]
0.02	10.000	7.154	**7.646**	**10.317**	11.985	[1.012, 1.459]
0.01	14.142	**11.118**	11.043	**14.415**	16.950	[1.112, 1.442]
0.001	44.721	**41.215**	35.605	**44.861**	53.601	[1.303, 1.419]
10^{-4}	141.421	**137.536**	112.813	**141.482**	169.500	[1.375, 1.415]
10^{-6}	1414.214	**1407.457**	1128.377	**1414.223**	1695.000	[1.407, 1.414]

TABLE 1. Numerical values of the bounds (2.2) and (2.1) for $M_H = \mathbb{E}\sup_{[0,1]} B^H$, together with the lower bound of [7, Cor. 2(i)] and the upper bound $1.695/\sqrt{H}$ of [14, Cor. 2 and Rem. 5]. (When $2^{2/H}$ is not an integer, $1.695/\sqrt{H}$ is smaller than the bound obtained in [14].) Bold entries mark the better bound in each pair. The last column gives $\sqrt{H}$ times our two bounds; these intervals contain $\sqrt{2} = 1.41421\ldots$

Grid maxima. Table 2 reports Monte Carlo estimates of $\mathbb{E}\max_{\Gamma_n} B^H$, based on exact samples of fractional Gaussian noise on Γ_n. The samples are generated by the Davies–Harte circulant embedding [30, 77, 38], which is exact for fractional Gaussian noise with $H < \frac{1}{2}$ because its autocovariances at non-zero lags are negative [28]. We use 20000 samples for $n \le 2^{14}$ and 2000 samples for $n \ge 2^{18}$, and the column "iid" gives the standard deviation of the maximum of n independent $N(0, \frac{1}{2})$ variables. We observe the following.

(a) All estimates lie below the bound $\mu_n/\sqrt{2}$ of Proposition 2.12, and they are close to it when $\kappa = H \log n$ is small. This is the regime of [15].

(b) For each fixed n, the ratio $\sqrt{H}\,\widehat{\mathbb{E}}\max/\Lambda(\kappa)$ increases as H decreases, towards $\mu_n/\sqrt{2\log n}$, its value for independent variables. Note that $\kappa = H\log n$ decreases with H, so this mixes the two parameters. At $H = 0.01$ the relative deficit $1 - \sqrt{H}\,\widehat{\mathbb{E}}\max/\Lambda(\kappa)$ is 15% for $n = 2^{10}$ and 9% for $n = 2^{21}$. This is comparable to the deficit $1 - \mu_n/\sqrt{2\log n}$ of the Gaussian maximum itself (13% and 7%), i.e. to the usual $\log\log n/\log n$ correction. In our data, which cover the four grid sizes of Figure 1(b) (the rows for $n = 2^{14}$ and 2^{18} are in the supplementary data), the deficit is below 10% only for $H = 0.01$ and $n \geq 2^{18}$, all in the phase $\kappa < \frac{1}{2}$.

(c) By Theorem 2.13, in the joint limit $H \downarrow 0$, $H\log n \to \kappa$, the grid captures only the fraction $\Lambda(\kappa)/\sqrt{2} < 1$ of M_H. This is a statement about the limit. At fixed H the realised fraction can be quite different: for $H = 0.1$ and $n = 2^{21}$ the limit gives $\Lambda(1.456)/\sqrt{2} = 0.81$, whereas Tables 1 and 2 only give $0.52 \lesssim \mathbb{E}\max_{\Gamma_n} B^H/M_H \lesssim 0.88$, from 2.583/4.935 and 2.583/2.940.

(d) The standard deviation of the grid maximum lies between 0.66 and 0.75, slightly above the standard deviation $(2H+2)^{-1/2} \in [0.645, 0.704]$ of the common mode $\int_0^1 B^H$ (Lemma 7.5). For the two entries of Table 2 with $H = 0.01$, its square agrees to within 2% with the sum of $(2H+2)^{-1}$ and the sample variance of $\max_{\Gamma_n} B^H - n^{-1}\sum_{\Gamma_n} B^H$. The standard deviation of this centred maximum is between 0.17 and 0.28, and for $H \leq 0.02$ it agrees with the i.i.d. benchmark to within 0.015. Its decrease with n is therefore the superconcentration of maxima of independent Gaussians, and not a feature of the hierarchy. Since Conjecture 9.2 concerns the regime $\kappa = \infty$, which no feasible grid reaches, these data neither support nor contradict it.

H	$\log_2 n$	κ	$\widehat{\mathbb{E}}\max_{\Gamma_n} B^H$	$\mu_n/\sqrt{2}$	$\sqrt{H}\,\widehat{\mathbb{E}}\max$	$\Lambda(\kappa)$	$\widehat{\mathrm{sd}}$(max)	$\widehat{\mathrm{sd}}$(max −avg)	iid
0.01	10	0.069	2.248 ± 0.005	2.297	0.225	0.263	0.741	0.242	0.248
0.01	21	0.146	3.481 ± 0.016	3.541	0.348	0.382	0.730	0.174	0.171
0.02	10	0.139	2.216 ± 0.005	2.297	0.313	0.372	0.738	0.243	0.248
0.02	21	0.291	3.426 ± 0.016	3.541	0.484	0.540	0.716	0.184	0.171
0.05	10	0.347	2.102 ± 0.005	2.297	0.470	0.589	0.735	0.249	0.248
0.05	21	0.728	3.088 ± 0.016	3.541	0.691	0.851	0.722	0.220	0.171
0.1	10	0.693	1.879 ± 0.005	2.297	0.594	0.831	0.716	0.259	0.248
0.1	21	1.456	2.583 ± 0.016	3.541	0.817	1.142	0.722	0.253	0.171
0.2	10	1.386	1.477 ± 0.005	2.297	0.661	1.123	0.681	0.272	0.248
0.2	21	2.911	1.773 ± 0.015	3.541	0.793	1.351	0.674	0.270	0.171

TABLE 2. Monte Carlo estimates (± one standard error) of grid maxima, compared with the bound of Proposition 2.12 and the limit $\Lambda(\kappa)$ of Theorem 2.13. The columns $\widehat{\mathrm{sd}}$ are the sample standard deviations of $\max_{\Gamma_n} B^H$ and of $\max_{\Gamma_n} B^H - n^{-1}\sum_{\Gamma_n} B^H$, and "iid" is the standard deviation of the maximum of n independent $N(0, \frac{1}{2})$ variables.

The Python code that produces Tables 1 and 2 and Figure 1 is provided as supplementary material. The Monte Carlo data are produced by `python mc_grid.py 2000 mc_grid.csv`, with the random seed 20261024 and the sample sizes stated above, and the tables and plot data by `python make_tables.py` and `python export_plot_data.py`.

9. Discussion and open problems

9.1. **Second-order behaviour.** By Theorem 2.1, $M_H - \sqrt{2/H} \le O(\sqrt{H}\log(1/H))$, so the limit superior of $M_H - \sqrt{2/H}$ as $H \downarrow 0$ is at most zero. The gap $O(\sqrt{\log(1/H)})$ in the lower bound comes from the greedy strategy, which loses the usual $\log\log b/\log b$ correction of Gaussian maxima at each level of the minorant tree. A comparison with branching Brownian motion suggests that the true correction is much smaller. For branching Brownian motion over a time horizon T with strictly decreasing variance $\sigma^2(t/T)$, Maillard and Zeitouni [58] showed that the maximum is

$$v_\sigma T - w_\sigma T^{1/3} - \sigma(1)\log T + O_{\mathbb{P}}(1),$$

and that the remainder converges in law, where w_σ is an explicit integral involving the largest zero $-\alpha_1 \approx -2.338$ of the Airy function; see [43, 61] for the order $T^{1/3}$ and its discrete-time analogue. In the normalisation of [58], the hierarchy of Section 1.2 is a branching Brownian motion on the infinite horizon $t \in [0,\infty)$, with time scale $T = 1/H$ and variance $\frac{H}{2}e^{-t/T}$ at time t. The integral defining w_σ then converges, the logarithmic term vanishes because $\sigma(\infty) = 0$, and a formal transfer gives $\sqrt{2/H} - M_H \approx (\alpha_1/\sqrt{2})\, H^{1/6}$.

Conjecture 9.1. $\lim_{H\downarrow 0}\big(M_H - \sqrt{2/H}\big) = 0$. *More precisely,* $\sqrt{2/H} - M_H$ *is positive and of order* $H^{1/6}$ *as* $H \downarrow 0$.

The transfer is only heuristic, for four reasons. In [58] the horizon is finite and σ is bounded below, whereas here the scales extend to infinity and their variances decay to zero. The geometry of fBm is not hierarchical. The comparison arguments of this paper lose more than the $H^{1/6}$ correction. Finally, fBm has a common mode; it does not affect M_H, but it has to be removed before the comparison with branching Brownian motion applies (Section 1.2(a)). Proving the conjecture would require controlling the non-hierarchical geometry of fBm at the level of the Airy corrections of [58], which seems to be a hard problem.

9.2. **Fluctuations of the maximum.** Write $\sup B^H = \bar{B} + \sup(B^H - \bar{B})$ with $\bar{B} = \int_0^1 B^H$. By Lemma 7.5, $\bar{B} \sim N(0, \frac{1}{2H+2})$. The centred field $B^H - \bar{B}$ has the hierarchical structure of Section 1.2 with no common mode, and maxima of hierarchical fields with many levels of small variance are *superconcentrated* [26]. In the branching analogue, the centred maximum of branching Brownian motion with decreasing variance converges in law [58]. Its fluctuations are therefore of order one in branching units, which carry the amplitude $\sqrt{H/2}$ here.

Conjecture 9.2. $\mathrm{Var}\big(\sup_{[0,1]}(B^H - \bar{B})\big) \to 0$ *as* $H \downarrow 0$. *Consequently* $\sup_{[0,1]} B^H - M_H \Rightarrow N(0, \frac{1}{2})$.

Borovkov and Zhitlukhin [15] proved that $\max_{\Gamma_n} B^H$ is approximately $N(\sqrt{\log n}, \frac{1}{2})$ when $H(\log n)^2 \to 0$. This is the analogue of Conjecture 9.2 at the other end of the phase diagram of Theorem 2.13, where the hierarchy is invisible. As explained in Section 8(d), simulations on feasible grids cannot test the conjecture.

9.3. **Gibbs measures and replica symmetry breaking.** For $\beta > \sqrt{2}$, Theorem 2.2 places $H\log Z_H(\beta)$ in a frozen phase. Consider the Gibbs measure $\mathcal{G}_{\beta,H}(dt) \propto e^{\beta B_t^H/\sqrt{H}}dt$ and the overlap $\mathsf{q} := H\log(1/|\tau-\tau'|)$ of two independent samples τ, τ' from $\mathcal{G}_{\beta,H}$. The CREM analogue is the overlap distribution of [21, Thm 3.6], and the analogues for the two-dimensional discrete Gaussian free field and for its scale-inhomogeneous version are [6] and [66]; see also [34, 70]. With the dictionary of Remark 2.8, these results predict continuous replica symmetry breaking with Parisi order parameter $x(y) = 2\sqrt{d}/(\beta\sqrt{g(y)})$ at log-scale y, which for fBm equals $\sqrt{2}e^y/\beta$.

Conjecture 9.3. *For $\beta > \sqrt{2}$, q converges in law (under the averaged Gibbs measure) to the distribution with distribution function $y \mapsto \min\bigl(1, \sqrt{2}\, e^{y}/\beta\bigr)$, $y \geq 0$. This law has an atom of mass $\sqrt{2}/\beta$ at 0 and a density on $(0, \log(\beta/\sqrt{2}))$. For $\beta < \sqrt{2}$, $\mathsf{q} \to 0$ in probability.*

The established route to such statements perturbs the variance on a window of scales and differentiates the free energy with respect to the perturbation [21, Lemma 3.5 and Thm 3.6]; for non-hierarchical fields this was carried out in [5, 66], together with the extended Ghirlanda–Guerra identities [67]. Theorem 2.7(ii) supplies the free energies of perturbed profiles. For example, adding to B^H an independent fBm of Hurst index Hh and small amplitude, as in Corollary 2.11(c), perturbs Φ while keeping it convex. What remains is to justify the differentiation, which requires convexity in the perturbation parameter and control of the overlap at the scales where the perturbation acts.

9.4. **Non-convex profiles.** If Φ is decreasing but not convex, one expects Φ to be replaced by its greatest convex minorant $\widehat{\Phi}$. For ultrametric index sets this is the concave-hull theorem of Bovier and Kurkova [20, 21], written there for the cumulative variance A.

Conjecture 9.4. *Let (X^{ε}) satisfy Assumption 2.5 with a continuous non-increasing function Φ, $\Phi(\infty) = 0$, in place of a profile. Then Theorem 2.7(i) holds with I_{Φ} replaced by $\int_0^{\infty} \sqrt{-\widehat{\Phi}'}$, whenever this integral is finite.*

For the upper bound, Lemma 4.1 allows arbitrary branching numbers, so one can try to group the scales of the majorant along the intervals where Φ differs from $\widehat{\Phi}$. Care is needed when there are infinitely many such intervals, or when $\widehat{\Phi}$ is not C^1. The lower bound would combine the greedy argument on the convex part with the sub-chord estimate of Lemma 7.2 on each block, where A lies below its concave hull (that is, $\Phi > \widehat{\Phi}$), as in the weak-correlation regime of variable-speed branching Brownian motion [19] and in the proof of Theorem 2.13. There the grid creates exactly such a non-convex effective profile.

9.5. **Further questions.**

(i) *Location of the maximiser.* Increments of fBm are stationary, and away from $t = 0$ the variance t^{2H} is close to 1 (for $H \log(1/t) \ll 1$), so the hierarchy is approximately translation invariant. We expect the maximiser of B^H on $[0, 1]$ to become uniformly distributed as $H \downarrow 0$, in contrast with the arcsine law at $H = \frac{1}{2}$.

(ii) *Pickands' constants.* Corollary 2.16 and Harper's bound [51] leave a gap in the exponential rate of $\mathcal{H}_{2H}$. Since $\mathcal{H}_{2H}$ is governed by the tail $\mathbb{P}(\sup B^H > u)$ as $u \to \infty$ rather than by M_H, a sharp result would need a large-deviation version of the present multiscale analysis.

(iii) *Multiplicative chaos.* In dimension one, Neuman and Rosenbaum [65, Cor. 2.2] proved that suitably recentred exponentials of $B^H/\sqrt{H}$ converge to a Gaussian multiplicative chaos in the whole subcritical range $\gamma < \sqrt{2}$. For fractional Brownian fields in dimension d, Hager and Neuman [49] prove convergence for $\gamma \leq \gamma^*(d)$, where $\gamma^*(d) > \sqrt{7d/4}$ falls short of the critical value $\sqrt{2d}$. The near-critical range for $d \geq 2$, the critical case and the frozen phase $\beta > \beta_c$ remain open; in the frozen phase Theorem 2.2 identifies the free energy, and log-correlated potentials are known to freeze [57].

(iv) *Designs.* By Remark 2.15, no deterministic set of n points captures a fraction θ of $\sqrt{2/H}$ unless $\log n \geq 2\theta^2/H$, and uniform grids attain this only for $\theta \leq \frac{1}{2}$. Can non-uniform deterministic designs, or designs that are refined adaptively along the sampled path, capture a fraction $\theta > \frac{1}{2}$ with fewer points than uniform grids?

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